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On the Expansion of Risk Pooling (Appendix)

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Abstract. Risk pooling has become an increasingly critical tool for managing risks among corporations, institutions, states, and nations, with examples including multinational pooling, decentralized insurance schemes, catastrophe risk pooling, and burden sharing for nuclear accidents. While there has been a rich literature on such practices, little is known from a theoretical viewpoint regarding the operational strategies of risk pools and in particular on issues about the effect of a pool's expansion on each existing member's welfare. This paper is the first to explore these issues by establishing different notions of consensus for the expansion of a risk pool: strong consensus, where both existing members and new candidates improve their risk measures due to the pool's expansion, and weak consensus, which refers to the willingness of existing members to remain in the pool. Under optimal risk sharing for each pool, we show that its properties regarding expansion's effects depend strongly on the underlying pricing rule. For instance, only with risk-adjusted equilibrium pricing are existing members willing to accept highly risky new members, whereas simple linear pricing excludes such members from the pool. Additionally, the impact of exogenous reinsurance on consensus is analyzed under both pricing rules. The established framework offers valuable managerial insights and policy implications, which we illustrate through two indicative case studies based on real data.

A. Caribbean Catastrophe Risk Insurance Facility

The Caribbean Catastrophe Risk Insurance Facility (CCRIF), established in 2007, is a regional insurance mechanism designed to provide Caribbean and Central American governments with parametric insurance coverage against natural disasters. CCRIF serves 19 Caribbean and 4 Central American countries, making it a critical risk management tool for nations highly vulnerable to climate-related events and natural disasters. Initially offering coverage for tropical cyclones and earthquakes, CCRIF expanded in 2013 to include excess rainfall. In 2020, it further extended its services to cover public utilities, specifically electric and water infrastructure. The key characteristics are:

- (i) **Parametric Insurance:** Unlike traditional insurance, CCRIF payouts are based on specific parameters or triggers (such as wind speed, rainfall, or earthquake magnitude) rather than actual assessed damage. This allows for rapid payouts within 14 days of an event.

- (ii) Pooled Risk: By pooling the risks of multiple countries, CCRIF can offer lower insurance premiums compared to what individual nations would pay on the global insurance market.
- (iii) Aggregate Deductible Cover (ADC): Introduced in 2017, ADC supports CCRIF member countries when modeled losses fall below the policy attachment point, yet actual on-the-ground losses are evident. It ensures financial assistance in cases where losses, though substantial, do not meet the threshold for standard coverage payouts. By bridging the gap between real-world damage and modeled loss thresholds, ADC offers additional financial protection, addressing discrepancies that may arise between theoretical and actual impacts.

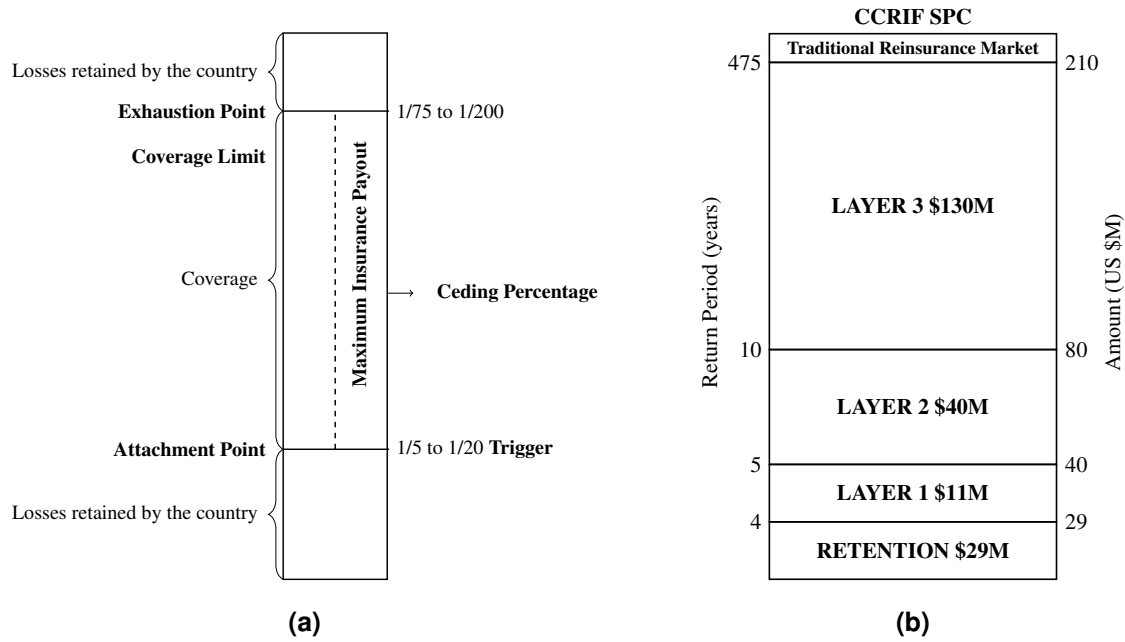
An example of ADC occurred in October 2018 when Haiti became the first member country to receive a payout (Bollmann and Wang (2019)). ADC is an optional feature that must be specifically selected when purchasing a policy. A payout is triggered if the modeled loss falls between 50% and 99% of the attachment point. Additionally, if the modeled loss is between 10% and 49%, a Disaster Alert from the ReliefWeb website must be issued for the country for an ADC payment to be made. The maximum payout a country can receive under ADC after an event is equivalent to the net premium paid for its tropical cyclone or earthquake policy (CCRIF SPC (2024b)).

- (iv) Reinstatement of Sum Insured (RSIC): RSIC was introduced alongside the ADC during the 2017–2018 policy year. It is an additional feature to ensure continued coverage after the initial policy limit is exhausted. It allows countries to access a “reinstatement of cover provision” once their maximum coverage limit under the initial policy has been fully utilized due to a catastrophic event. This feature is particularly useful in scenarios where multiple disasters occur within a short time-frame (Bollmann and Wang (2019)). For instance, in 2021, following a significant earthquake, Haiti received a payout that exhausted its earthquake policy limit. Thanks to RSIC, Haiti’s policy was reinstated, providing continued coverage for the remainder of the policy year (CCRIF SPC (2021a)). RSIC is also an optional feature like ADC; member countries need to decide whether they want to include RSIC as part of their policy when they sign the insurance agreement. After a payout that depletes the policy’s coverage limit, RSIC automatically reinstates the coverage to its original amount, maintaining the same terms and conditions as the initial policy. The premium for the reinstated coverage is typically a fraction of the original policy premium. This cost is predetermined and outlined in the policy agreement. (CCRIF SPC (2024a))

A.1. Insurance Module: Tropical Cyclone

As stated in the main body of the paper, in the first case study we consider only tropical cyclones (mainly due to the lack of data). Tropical Cyclone (TC) insurance module calculates payouts to each insured country based on modelled loss estimates following a TC event. A country’s policy is activated only if the modelled loss from a TC event meets or exceeds the attachment point, meaning no payout is made for losses below this threshold.

Figure 1 Insurance module and the aggregate excess of loss program for the tropical cyclone.



CCRIF's insurance module is presented in Figure 1a (CCRIF SPC (2023b)). Each country's insurance policy parameters are carefully selected to provide optimal coverage that aligns with its risk management strategy while staying within budget. The trigger point range of 1/5 to 1/20 represents the likelihood or frequency threshold at which an insurance policy begins to pay out. The maximum payout a country can receive is determined by subtracting the attachment point from the exhaustion point and multiplying the result by the ceding percentage. The attachment point is the loss level at which the policy begins to pay out, while the exhaustion point represents the upper threshold beyond which no further payouts are made. The coverage limit, which is the highest dollar amount a policy can pay out, is based on the range between these two points and adjusted by the ceding percentage. The ceding percentage plays a crucial role in determining the premium. Once the attachment and exhaustion points are set, there is a direct relationship between the premium amount and the ceding percentage. A higher ceding percentage results in a higher premium, as it increases the portion of potential losses that can be covered.

It is important to mention that CCRIF insurance exclusively covers government losses, which are calculated as a percentage of a country's total national losses. This structure ensures that payouts are aligned with the financial impact on public sector infrastructure and services. It is therefore reasonable to assume that the payout for each country is analogous to its GDP, a method that we shall apply for calibrating the models' parameters.

CCRIF retains US \$29.0 million for payouts and has secured an additional US \$181.0 million in reinsurance capacity above this retention to enhance its claims-paying capacity (CCRIF SPC (2023a)). This

reinsurance structure, capped at US \$210.0 million, provides coverage for aggregate annual losses (as in our framework). This structure is illustrated in Figure 1b.

A.2. Model Calibration

CCRIF uses a parametric insurance model that follows the same general structure across all member countries, but the parameters and triggers within that model can vary based on the individual contracts signed by each country. The parameters and thresholds (such as wind speed, rainfall intensity, or earthquake magnitude that trigger payouts) are customized based on each country's risk profile and needs. Since this information is not publicly available, it is difficult to accurately estimate the actual losses based solely on their payout data. The model triggers payouts based on predefined parameters (such as wind speed or rainfall), rather than on detailed assessments of actual damage, making it challenging to directly correlate payouts with real losses.

The yearly payout data is considered as loss data, which is divided by 100,000 for the following calculations. For each year the realization of the sum of losses is denoted by $S_{n,year} = \sum_{i=1}^n X_{i,year}$, where $X_{i,year}$ is the individual loss for country i in any year of the data. In order to estimate the annual risk variance for each country, we first estimate the mean and the variance of the aggregate annual losses (m_n, s_n^2) . The parameters of m_n and s_n^2 are found through the maximum likelihood estimate. The maximum likelihood estimate (M USD) is given by

$$\hat{m}_n = \frac{1}{n} \sum_{year} S_{n,year} \approx 159.56, \quad \hat{s}_n^2 = \frac{1}{\#year} \sum_{year} (S_{n,year} - \hat{m}_n)^2 \approx 56,598.28$$

Based on this, we find the mean and the variance of each country j using the following approximation:

$$\text{Adjusted_Variance}_i = \hat{\sigma}_i^2 := k^2 \times (\text{Occurrence_Probability}_i) \times (1 - \text{Occurrence_Probability}_i) \times (\text{GDP}_i)^2$$

where k is the constant to match the sum of adjusted variances to the variance of total loss, under the assumption that we have the constant correlation $\rho = 0.3$ among any pair of two countries, *i.e.*,

$$\sum_j \hat{\sigma}_j^2 + 2\rho \sum_{i \neq j} \hat{\sigma}_i \hat{\sigma}_j = \hat{s}_n^2$$

Then, $k \approx 0.000386$. Based on this, the adjusted variance can be computed, which is presented in Table 1.

Given Table 1, by approximating the payouts as the actual losses, we can apply our model to evaluate the potential impact of a new candidate joining in the following year. For the implementation, we need to estimate two key parameters: (1) risk tolerance coefficients, and (2) correlation coefficients among countries' risks. To simplify the analysis, it is assumed that all countries have the same pairwise correlation $\rho = 0.3$, and the same risk tolerance level $\delta = 1$. Note that higher correlations are expected among some countries that frequently experience losses from same disasters.

Table 1 Summary Table for each member.

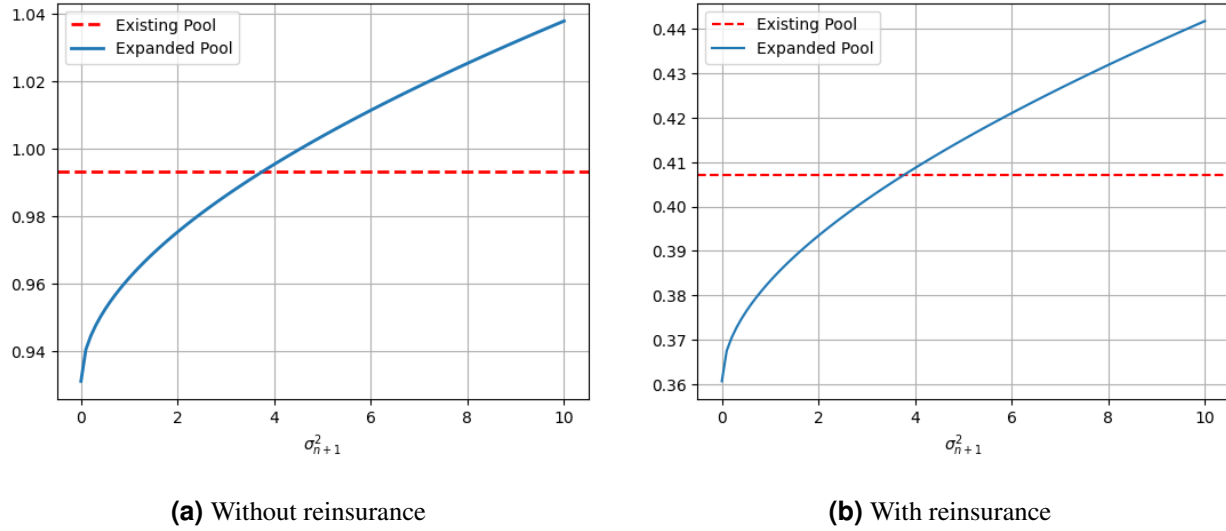
Members	Year	GDP (\$M USD)	Occurrence Probability	Adjusted Variance
Anguilla	2007	452	0.0727	0.016
Antigua & Barbuda	2007	2,030	0.0364	0.081
Barbados	2007	6,390	0.1091	7.254
Belize	2007	3,280	0.0545	0.478
Cayman Islands	2007	6,990	0.0182	0.241
Dominica	2007	654	0.0364	0.008
Grenada	2007	1,320	0.0910	0.215
Haiti	2007	14,340	0.0545	9.133
Jamaica	2007	32,420	0.0545	46.680
Saint Lucia	2007	2,520	0.0545	0.282
St. Kitts & Nevis	2007	1,080	0.0364	0.023
St. Vincent & the Grenadines	2007	1,070	0.0727	2.90
The Bahamas	2007	14,340	0.0545	0.090
The Republic of Trinidad and Tobago	2007	28,140	0.0364	15.630
Turks & Caicos Islands	2007	1,400	0.0727	0.155
Nicaragua	2015	14,340	0.0910	25.369
British Virgin Islands	2018	476	0.0182	0.001
Panama	2018	83,380	0.0182	34.307
Guatemala	2019	247,610	0.0182	302.549

Following its history, the existing pool consists of the following 16 founding members- Dominica, Saint Lucia, Turks & Caicos Islands, Haiti, Anguilla, Barbados, St. Vincent & the Grenadines, St. Kitts & Nevis, Belize, Antigua & Barbuda, The Bahamas, The Republic of Trinidad and Tobago, Jamaica, Grenada, Cayman Islands. We consider the joining of other members- British Virgin Islands, Nicaragua, Panama, and Guatemala. The evolution of membership is summarized in Table 1 (CCRIF SPC (2025), Caribbean Community (2025), World Bank (2015), CCRIF SPC (2020), and CCRIF SPC (2021b)).

A.3. Results

As described in the paper, the expansion process is considered one candidate at a time. A candidate may represent a single country or a region consisting of multiple countries. To simplify the analysis, we initially focus on the expansion involving a single country. The simulation results are shown in Figures 2 and 3, and the summary of expansion is presented in Table 2.

The model suggests that it might not always be advantageous to incorporate new members. The addition of members with high variance, such as Guatemala, Nicaragua, and Panama—each exhibiting significantly higher variances than the existing pool—could have an adverse impact on the pool under risk-conservative pricing rules like Fairness Principle (Figure 2a). In Figure 2b, this holds true regardless of whether reinsurance is in place. Conversely, if members' decisions are made based on risk-seeking pricing rules like Equilibrium Principle, the British Virgin Islands, Nicaragua, and Panama would be unsuitable for inclusion due to its low risk profile (Figure 3a). In Figure 3b, under the stricter acceptance criteria with reinsurance,

Figure 2 CCRIF: Expansion under Fairness Principle without and with reinsurance.

Note: On the left panel, the deviation-tolerance ratios are shown, while the values of l function (13) from the main text are presented on the right panel. In both cases, the existing members benefit from the expansion when blue lines are below red lines.

Table 2 CCRIF: Summary of Expansion (checkmark means SC)

	DTR		W/O Reinsurance				W/ Reinsurance			
			FP		EP		FP		EP	
pairwise correlation	0.1	0.3	0.1	0.3	0.1	0.3	0.1	0.3	0.1	0.3
Existing Pool	1.02	0.99	-	-	-	-	-	-	-	-
British Virgin Islands	0.04	0.03	✓	✓			✓	✓		
Guatemala	23.04	17.39			✓	✓			✓	
Nicaragua	6.67	5.03			✓				✓	
Panama	7.76	5.85			✓				✓	

even Guatemala is not accepted. In this context, the pool favors candidates seeking to offload risks rather than those inclined to assume additional risks.

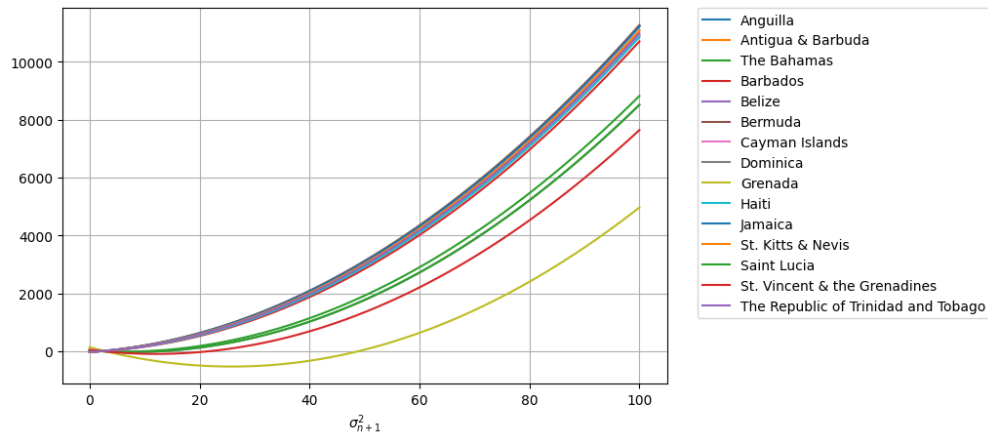
B. Potential Healthcare Sharing in China

This example explores the application of fairness and equilibrium principles with and without reinsurance, using health insurance data in China. The first section provides an overview of the data used in the analysis, and the second section explains the design of the simulation setup. The key findings of the analysis are summarized in the result section. It highlights how the conditions for expanding risk pools vary under the fairness and equilibrium principles, both with and without reinsurance, providing critical insights into the practical implications of each framework.

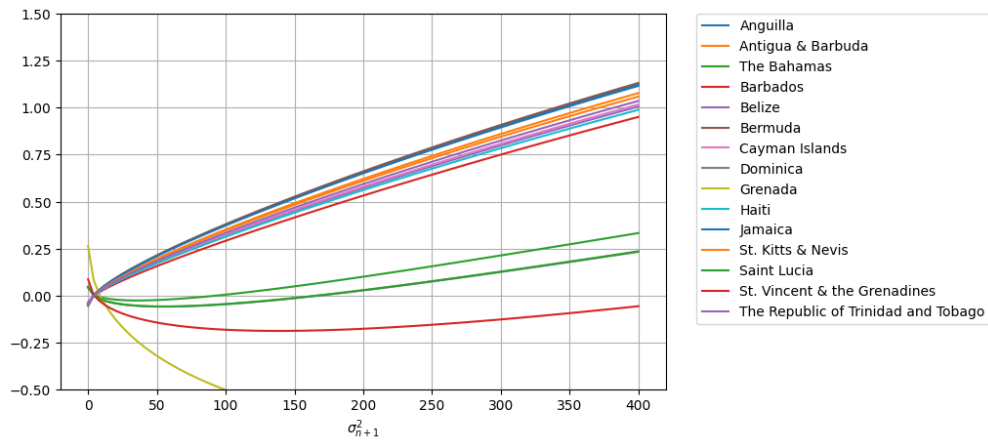
B.1. Data

The data is collected, pertaining to healthcare services and insurance claims in the year of 2020 within the framework of China's health insurance system. The dataset provides a range of information that includes

Figure 3 CCRIF: Expansion under Equilibrium Principle without and with reinsurance.



(a) Without reinsurance



(b) With reinsurance

Note: On the top panel, the values of f function (19) from the main text are presented, while the results of Monte-Carlo simulations for the difference in entropic risk measures are presented on the bottom panel. In both cases, each member in the existing pool benefits from the expansion when the graph is above zero.

demographic details, healthcare service utilization, and financial compensation for medical treatments. Specifically, the data includes the following attributes:

1. Demographic Information: Gender and date of birth. The birth date specifies the age of the individuals at the time of the claim, which can be used to infer age.
2. Healthcare Service Details: The names of hospitals visited are listed, along with the corresponding regions or districts in China.
3. Medical Condition: The specific medical condition or diagnosis for which the individual sought treatment is identified.
4. Financial Information: The amount of financial reimbursement or coverage provided under the insurance claim, denoted in yuan.

Table 3 Sample Data

Category	First person	Second person	Third person
Gender	M	F	F
Date of Birth	1981-03-17	1987-07-31	1978-09-30
Hospital Visited	Luodian Hospital, Baoshan District, Shanghai	Beijing Longxi Hospital	Shenzhen Longhua District Maternal and Child Health Hospital
Disease Name	Lower limb fracture	Thyroid Nodules	Missed abortion
Compensation Amount	198.00	185.94	9.90

Table 4 Mean and variance of compensation for each region

Region	Count	Mean	Variance
Beijing	4,357	1.78	11.90
Chengdu	312	0.88	3.67
Chongqing	400	1.40	9.68
Dalian	450	2.34	30.90
Guangzhou	223	2.40	31.81
Hangzhou	116	1.45	22.55
Nanjing	1,096	1.29	12.88
Qingdao	172	1.29	9.86
Shanghai	11,107	1.35	9.16
Shenyang	119	2.17	14.01
Shenzhen	233	1.69	9.67
Suzhou	214	1.29	19.56
Tianjin	1,211	1.06	15.57

The example data is given in Table 3.

B.2. Model Calibration

The data was organized by extracting regional information based on the hospitals visited by individuals for their medical treatment. The hospital names in the dataset either explicitly mentioned the name of the region, or provided enough contextual clues to infer the region.

After determining the region for each claim, the data was grouped accordingly. This grouping allowed us to find the estimates for the parameters of distributions across different areas. To standardize the compensation amounts for easier simulation, the original values were divided by 100. Then, for each region, we calculated the number of claims, the average mean compensation, and the variance of the compensation amounts. These estimations are listed in Table 4.

We selected regions with a sufficiently large number of claims and ensured that the chosen cities have comparable size. A common pairwise correlation of 0.3 and a common risk tolerance equal to 1 for all cities are assumed.

To construct the case study following our framework, we need to impose an existing group of cities. For this, we consider a reasonable formation of geographic criterion and let major cities of northeastern China, such as Beijing, Tianjin, Shenyang, and Dalian as the existing agents. We then examine the expansion of this pool for a number of different candidates (e.g. other Eastern Coastal cities such as Shanghai, Nanjing, Suzhou, and Hangzhou, or Southern Chinese cities like Shenzhen and Guangzhou).

Table 5 Summary of statistics and decision-making for the expansion of health-share

	D/T	W/O Reinsurance		W/ Reinsurance	
		Fairness	Equilibrium	Fairness	Equilibrium
Existing pool	2.9	-	-	-	-
Guangzhou	5.64		✓		✓
Chengdu	1.92	✓	✓	✓	
Chongqing	3.11	✓	✓	✓	
Qingdao	3.14	✓	✓	✓	
Shanghai	3.03	✓	✓	✓	
Shenzhen	3.11	✓	✓	✓	
Others	-	✓	✓	✓	✓

B.3. Results

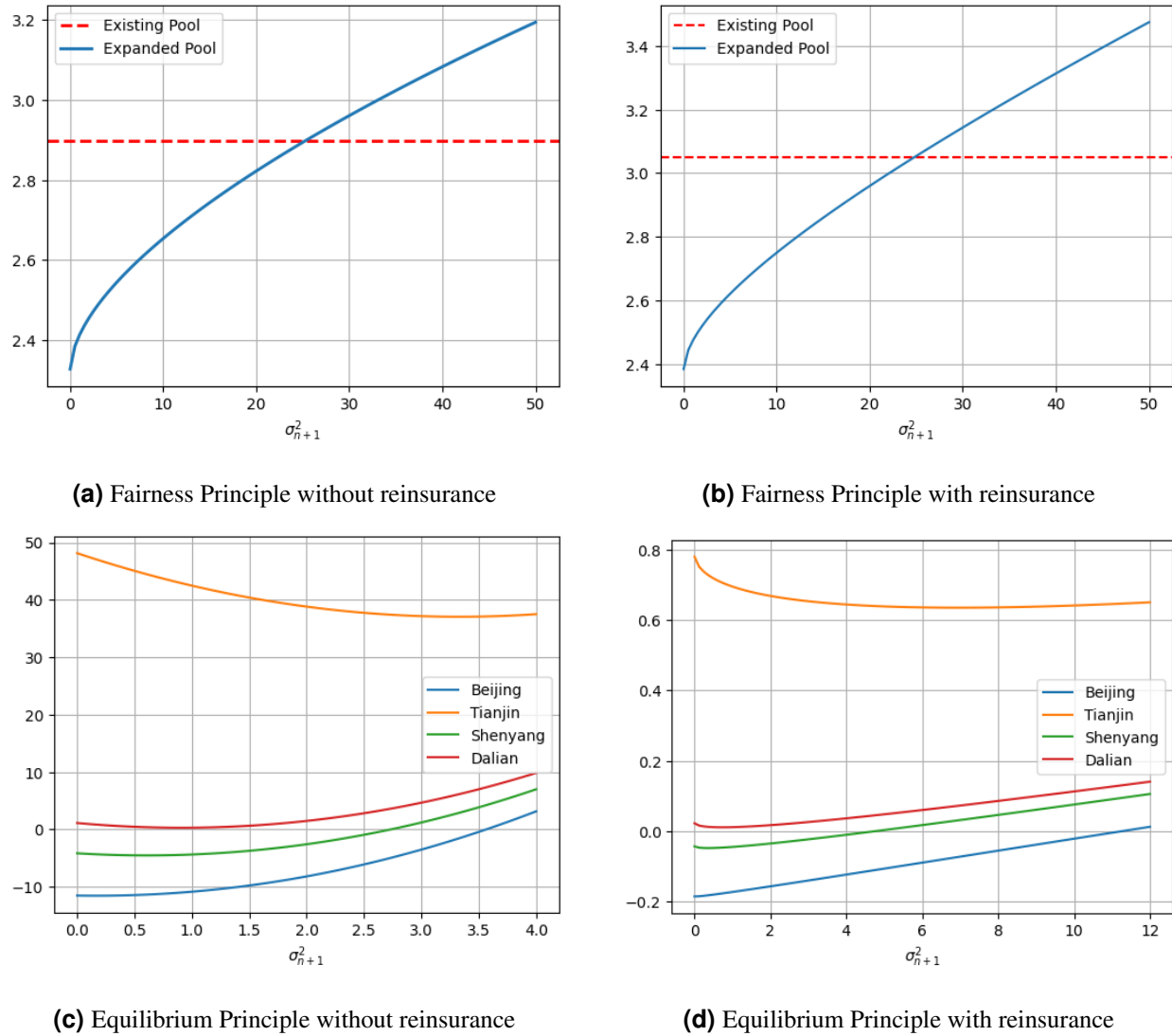
As outlined in the paper, the expansion considers one candidate, which could be a single city or a region comprising multiple cities. We work on the simplified case of single city expansion (where the reinsurance level β is set equal to 1.64).

The FP dictates that the expansion will be accepted in SC sense if the variance of the candidate city is less than 25, regardless of the presence of reinsurance (Figures 4a and 4b). Based on this criterion, the existing pool agrees to include all candidate cities except Guangzhou (whose risk's variance is rather high). In contrast, under EP, the acceptance threshold varies depending on whether reinsurance is applied. Without reinsurance, EP allows expansion to candidates with variances larger than 3.5, meaning that all cities are accepted (Figure 4c). However, with reinsurance, the threshold rises to 11.2, resulting in the exclusion of cities such as Chengdu (Figure 4d). The results are summarized in Table 5 and Figure 4.

These results align with the expectations outlined in the paper and highlight the fundamental differences between the fairness principle and the equilibrium principle in evaluating risk pool expansions. Under the fairness principle, lower-risk candidates are naturally favored because the fairness principle only charges premiums based on expected losses, which does not incorporate the additional compensation that higher-variance candidates could bring to the pool. Consequently, the fairness principle results in the acceptance of lower-risk candidates and excludes those exceeding the variance threshold, as seen in the rejection of Guangzhou. This preference remains unchanged regardless of whether reinsurance is applied, as the premium structure under fairness remains tied solely to the expectation of losses.

In contrast, the equilibrium principle reflects a preference for higher-risk candidates due to its pricing mechanism, which charges premiums based on variance. Candidates with higher variance are more attractive as they generate greater compensation for the existing members. Therefore, under the equilibrium principle without reinsurance, candidates with high variance are always preferred, as their higher risk translates into higher premiums that benefit the pool.

With the introduction of reinsurance under the equilibrium principle, the pool becomes less likely to accept candidates due to the competition between existing members and the reinsurer: the premiums are now partially transferred to the reinsurer. This reduces the incentive for the existing pool members to accept

Figure 4 Strong Consensus under Fairness and Equilibrium Principle with and without reinsurance.

Note: On the top-left panel, the deviation-tolerance ratios shown, while the values of l function (13) from the main text are presented on the top-right panel. In both cases, the existing members benefit from the expansion when the blue line is below the red line. On the bottom-left panel, the values of f function (19) from the main text are presented, while the results of Monte-Carlo simulations for the difference in entropic risk measures are presented on the bottom-right panel. In both cases, each member in the existing pool benefits from the expansion when the graph is above zero.

candidates, as the additional compensation from high risks is no longer fully retained within the pool. As a result, the equilibrium principle with reinsurance imposes a stricter acceptance threshold, excluding cities such as Chengdu, whereas without reinsurance, all candidates were accepted.

In summary, the results from two previous examples align with the theoretical expectations outlined in the paper, highlighting the distinct evaluation criteria of the two principles and the influence of reinsurance on the decision-making process. These findings also indirectly demonstrate why, in certain scenarios, the equilibrium principle can yield outcomes that are more beneficial from a societal perspective than the fair-

ness principle. For instance, in situations where the existing pool possesses substantial wealth and is willing to take higher risks for greater premiums, the equilibrium principle can produce more favorable results compared to the fairness principle.

C. Deviation of Normality Assumption

Our main analysis could be developed and studied under different probability distributions. Although it would be infeasible to have analytic conditions for weak and strong consensus, our setting provides all the necessary steps to simulate the conditions for both consensuses.

Of course, imposing other types of common risks' distributions, (such as Exponential, Pareto, and Gamma), will not produce the exact same conditions as in the case of the normal distribution. However, we could examine whether the choice of other distributions will give similar *qualitative* results. In other words, while the exact consensus conditions will not hold, we may get the same economic messages regarding, for instant, the effect of different pricing rules to acceptance of riskier candidates, the effect of higher risk correlations etc.

In order to examine the above, we have conducted a number of simulations in the view of our main results, under different common risk distributions and below we state a summary of the main outcomes. In short, we verify that even under heavy-tailed distributions our qualitative messages remain untouched.

More precisely, a central implication of this analysis is that risk pools tend to exhibit conservative behavior under the fairness principle, admitting only low-risk candidate. In contrast, under the equilibrium principle, existing members demonstrate a preference for high-risk candidates, as higher premiums offset the additional risk. This difference arises from the nature of the two pricing principles: while the fairness principle determines premiums based solely on the expected value of risk, the equilibrium principle incorporates also the second moment. Our simulation aims to examine whether this implication holds for other well-used distributions. We also could verify that higher correlation means that consensus conditions get stricter (since the diversification effect gets weaker).

We have run several Monte-Carlo simulations, considering a wide range of different parametrizations for Exponential, Pareto and Gamma distribution. Through this exercise we did verify our main qualitative results. In Section C.2- we give some indicative examples of these simulations.

C.1. Simulation of Risk Measures under Different Distributions

Suppose as in the main text that there are n agents in an existing risk-sharing pool. The random vector $\mathbf{X} = (X_i)_{i \leq n}$ denotes the risk allocation before the risk sharing (all realized at a terminal time), and $\mathbf{Y}^{(n)} = (Y_i^{(n)})_{i \leq n}$ stands for the risk allocation after the sharing, where the superscript (n) is used to indicate the size of the risk pool. Recall that the entropic risk measure of agent i , is a map ρ_i defined as

$$\rho_i[X] = \text{CE}_i[X] := \delta_i \log \mathbb{E}[e^{\frac{1}{\delta_i} X}],$$

where $\delta_i > 0$ is the risk tolerance of agent i , while for future reference we define $\Delta_n = \sum_{i=1}^n \delta_i$ as the aggregate risk tolerance of the existing (n -populated) group of agents. It is a well-known result that under cash-invariant risk measures Pareto-optimal allocations of X are determined by the following problem

$$\min \left\{ \sum_{i=1}^n \rho_i[Y_i^{(n)}] \right\}, \quad (1)$$

subject to the loss conservation condition $\sum_{i=1}^n X_i = \sum_{i=1}^n Y_i^{(n)}$. Define the relative risk tolerance $\lambda_{n,i} := \delta_i / \Delta_n$ and the aggregate risk $S_n := \sum_{i=1}^n X_i$. As mentioned before (see e.g. Barrieu and El Karoui (2005)), the linear risk-sharing rule is indeed the optimal even when the risks are not jointly normal. On the other hand, for the distributions whose moment-generating functions are not well-defined, the linear sharing could be considered as a meaningful “sub-optimal” sharing among agents in an approximating sense. Such sharing rule is given by

$$\mathbf{Y}^{(n)} = \mathbf{A}\mathbf{X} + \mathbf{p}^{(n)}, \quad (2)$$

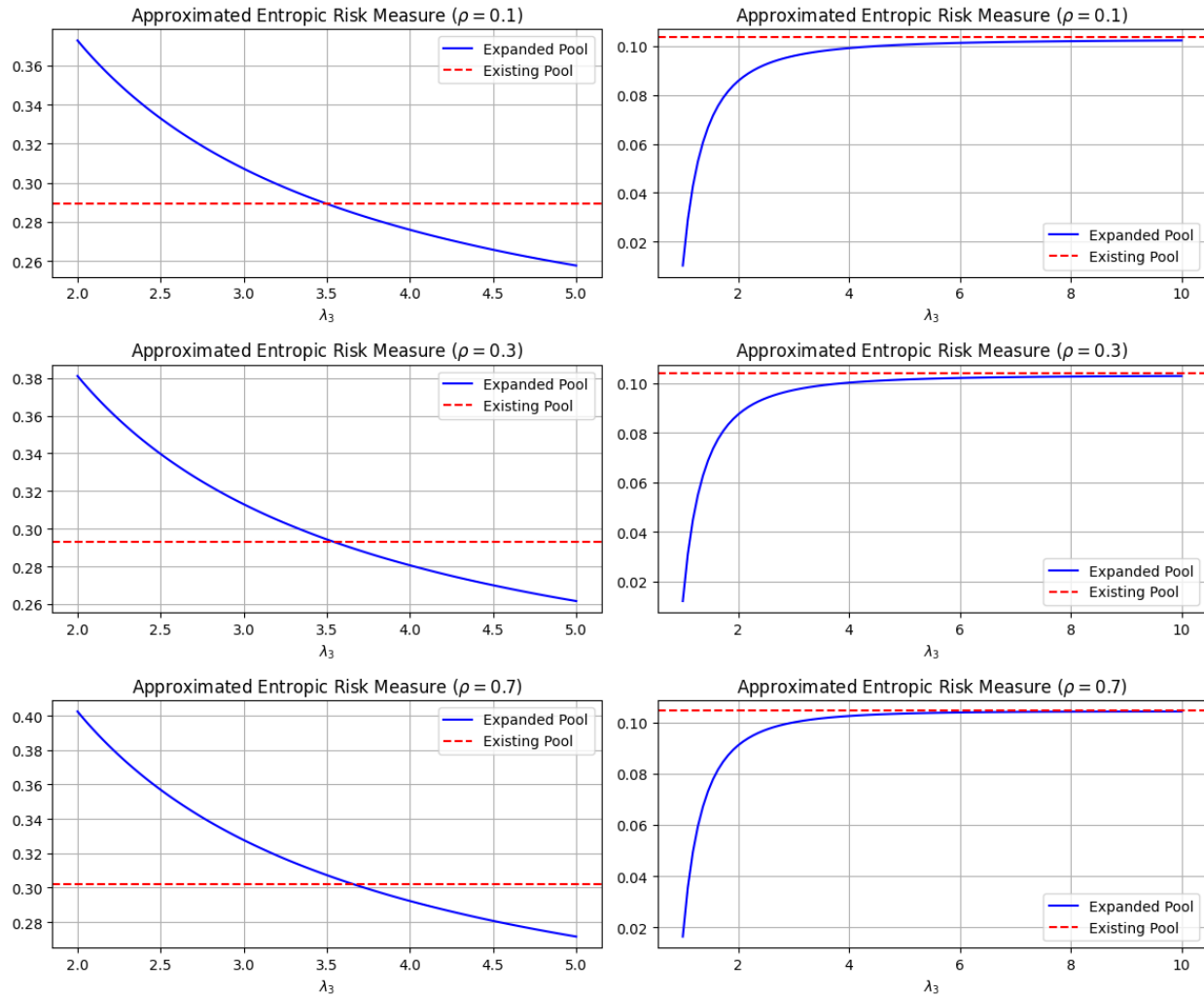
where $\mathbf{A} = (a_{i,j})_{i,j=1,\dots,n}$ such that $a_{i,j} = \lambda_{n,i}$ for all $i, j = 1, \dots, n$. The internal transfer $\mathbf{p}^{(n)} = (p_{n,i})_{i=1,\dots,n}$ within the group of n agents is n -dimensional vector that is fixed by the pricing rule such as fairness principle and equilibrium principle. In the following simulations, the expansion under the strong consensus is considered, especially from the perspective of existing members, which must satisfy

$$\rho_i(Y_i^{(n)}) \geq \rho_i(Y_i^{(n+1)}), \quad \forall i = 1, 2, \dots, n. \quad (3)$$

Note again that the simulation conducted in this section is an approximation. For a finite sample size, the moment-generating function of Pareto distribution is finite because the extreme tail values are underestimated in the simulation process. The moment-generating function, which theoretically diverges to infinity for any positive $t > 0$, is computed as a sample-based approximation: $\mathbb{E}[e^{tX}] \approx \frac{1}{N} \sum_{i=1}^N e^{tX_i}$. With a limited number of samples, it remains finite as the rare extreme tail events that dominate the divergence are unlikely to occur. However, as the sample size increases, the likelihood of including these extreme values grows, and the sample approximation begins to reflect the heavy-tailed nature of the Pareto distribution. In the limit as the sample size approaches infinity, the moment-generating function diverges, aligning with its theoretical behavior.

While the theoretical divergence of the moment-generating function is a characteristic of the Pareto distribution, the approximated results obtained through finite sample simulations are sufficient for our demonstration. The purpose of this simulation is to provide a consensus condition under other distributions than normal distributions. These approximations capture the key features of the Pareto distribution, including its heavy-tailed nature, without requiring an infinite number of samples. The results still allow us to analyze the impact of extreme values on risk measures, enabling us to demonstrate the consensus condition under the Pareto distribution. In the same manner, the correlated exponential and Gamma distributions have a finite moment generating function for a finite sample size which is again enough for the demonstration in this section.

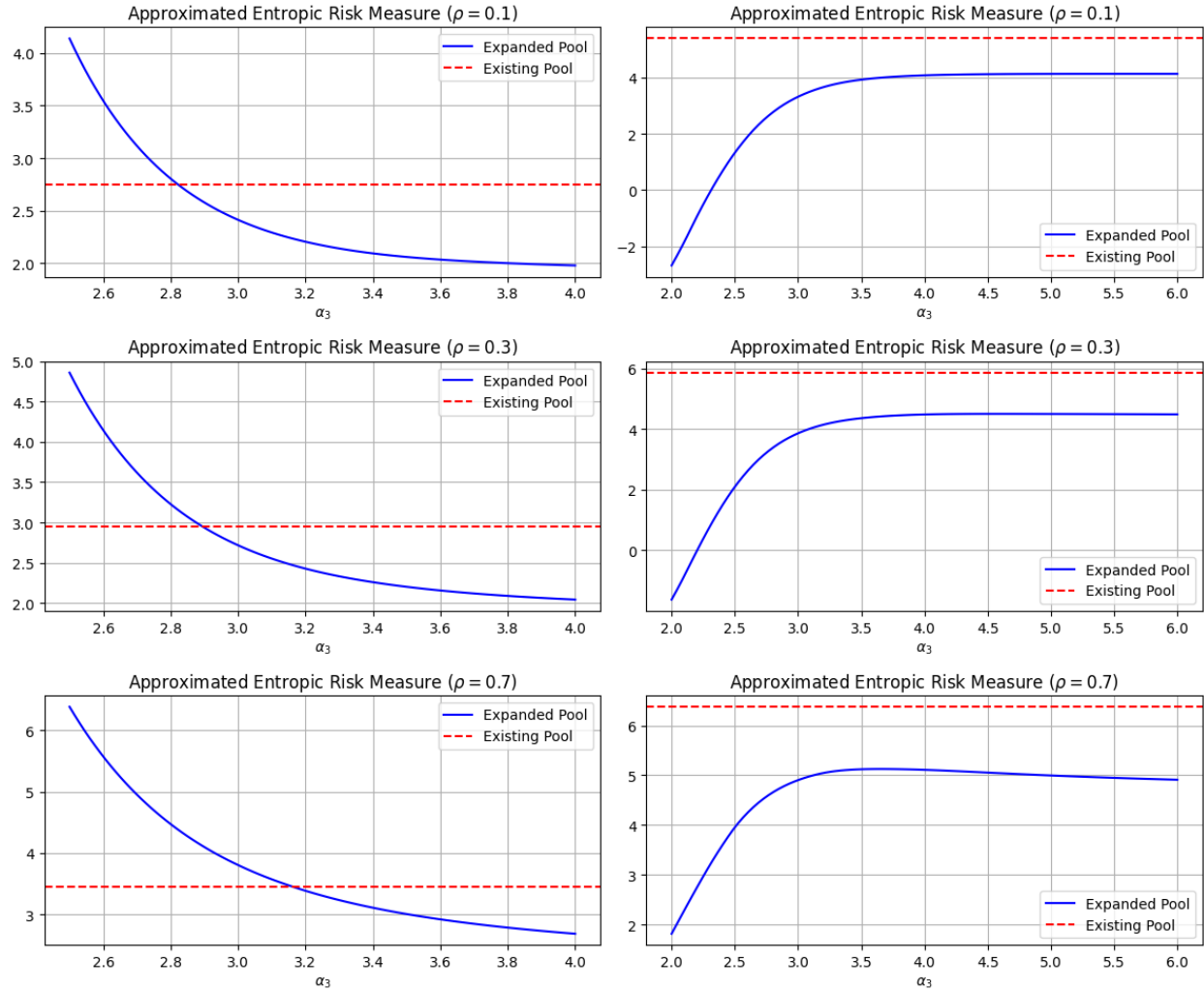
Figure 5 Finite sample approximations of risk measures for the existing and expanded pools, based on the exponential distribution, under the fairness and equilibrium principles with varying values of λ_3 .



C.2. Results

The presented simulations assume an existing risk pool comprising two members and consider the main case of common pair correlation ($\rho = 0.1, 0.3$, and 0.7) and common agents' risk tolerance (equal to one). Figure 5 presents the simulation results for exponential distributions: The loss distributions of the existing members and the candidate follow exponential distributions with parameters λ_1 , λ_2 , and λ_3 respectively. On the left panels in Figure 5 the fairness principle is imposed, where $\lambda_1 = 3$ and $\lambda_2 = 5$. Therein, the risk measure of the expanded pool is shown to decrease with respect to λ_3 confirming that candidates with lower variance are more likely to be accepted under the fairness principle. In contrast, on the right panels in Figure 5, the risk measure of the expanded pool increases with respect to λ_3 under the equilibrium principle, where $\lambda_1 = 10$ and $\lambda_2 = 15$. The observation that candidates with sufficiently high variance are always accepted

Figure 6 Finite sample approximations of risk measures for the existing and expanded pools, based on Pareto distribution, under the fairness and equilibrium principles with varying values of α_3 .

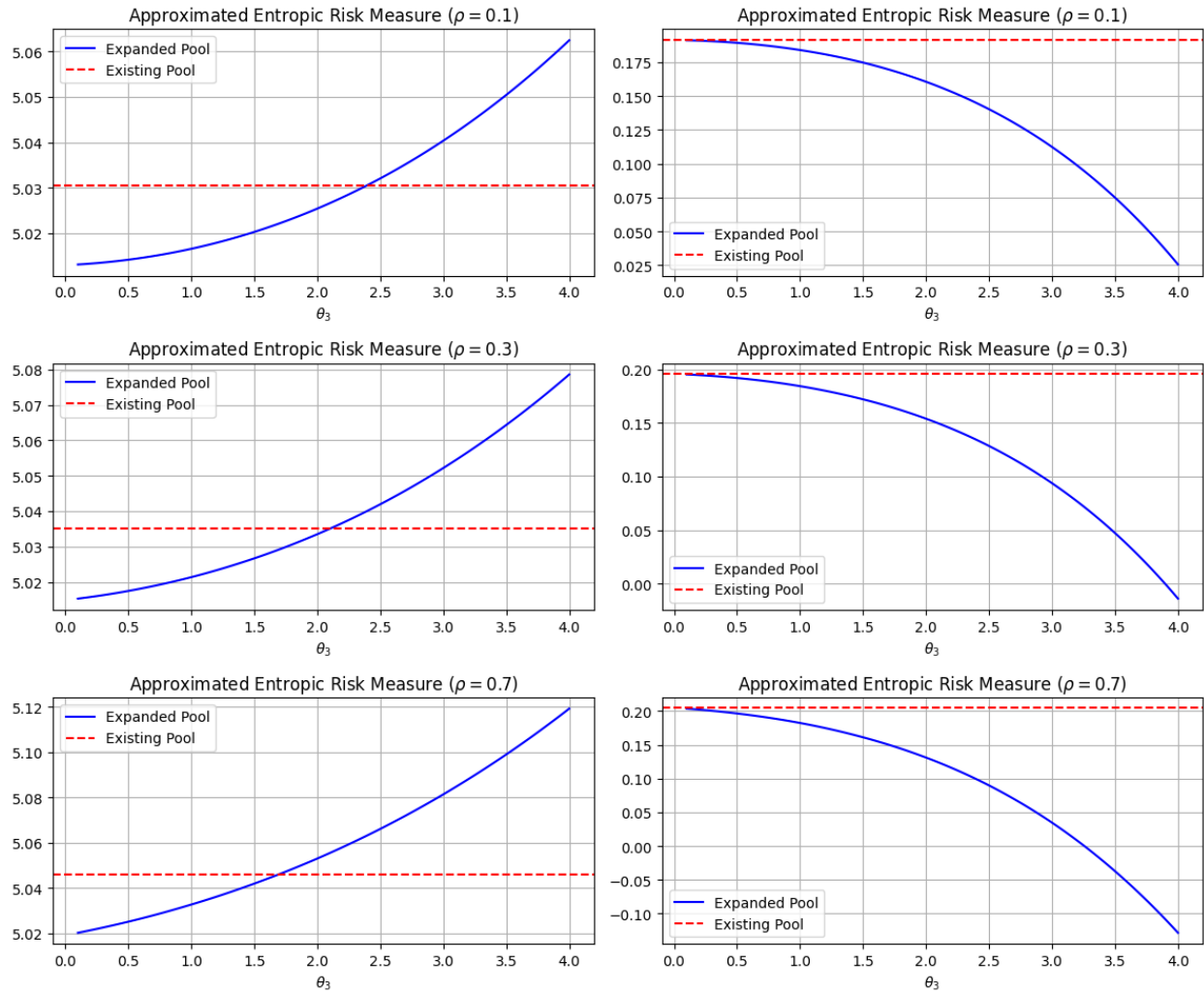


under this equilibrium pricing aligns with the analytic formulas under normally distributed risks. We also verify that higher correlation means higher risk measure under the expanded risk pool.

In Figure 6, the Pareto distribution is used for the simulation. Recall that the linear risk-sharing rule is indeed optimal for any bounded approximated version of the Pareto distribution. Hence, for any practical perspective the numerical simulations of agents' risk measures could be used to (at least) examine the qualitative take-home message of our work.

With the scale parameter fixed at 1, the shape parameters are given by $\alpha_1 = 3$, $\alpha_2 = 4$, and α_3 for the existing members and the candidate, respectively. In Figure 6, the results align with previous findings: candidates with lower risk are preferred under the fairness principle, whereas candidates with sufficiently high risk are consistently favored under the equilibrium principle. Notably, the upward slope followed by a

Figure 7 Finite sample approximations of risk measures for the existing and expanded pools, based on the Gamma distribution, under the fairness and equilibrium principles with varying values of θ_3 .



downward slope (on the right panels in Figure 6) illustrates the prediction from the main text, indicating the possibility of accepting candidates with either sufficiently low or sufficiently high risk under the equilibrium principle. The qualitative similarity of results obtained using the Pareto and normal distributions indicates that these findings are robust for heavy-tailed distributions as well.

Finally, Section 7 illustrates an example of our simulation under Gamma distribution. In this example, we assume a fixed shape parameter of 1, with scale parameters set as $\theta_1 = 1$, $\theta_2 = 2$, while θ_3 varies. Consistent with earlier cases, the simulations in Figure 7 confirm that under the fairness principle (on the left panels), candidates with lower risk are preferred, while under the equilibrium principle (on the right panels), candidates with higher risk are favored.

The simulation results demonstrated that the qualitative conclusions remain consistent across normal and other distributions, even when applied to settings that differ from the normal distribution. For instance, in the exponential distribution, the single parameter λ simultaneously affects both the mean and the variance, whereas in the normal distribution, variance is distribution's parameter. This consistency suggests that the conclusions are likely to extend to other distributions as well, despite the involved analytical complexities.

D. Supplementary Materials for Section 3

D.1. Proofs of Section 3

Proof of Theorem 2

Theorem 1 implies that risk-sharing of n agents is proportional and determined by $\lambda_{n,i}$. Specifically, the risk allocated to the agent i is

$$\lambda_{n,i}S_n \sim \text{Normal}(\lambda_{n,i}m_n, \lambda_{n,i}^2s_n^2),$$

where $s_{n+1}^2 = s_n^2 + \sigma_{n+1}^2 + 2\rho(S_n, X_{n+1})s_n\sigma_{n+1}$. By the fairness principle, the cash transfers of agent i in the pools with n and $n+1$ agents are

$$\pi_{n,i} = \lambda_{n,i}m_n - \mu_i;$$

$$\pi_{n+1,i} = \lambda_{n+1,i}m_{n+1} - \mu_i,$$

respectively. The entrance of the candidate benefits the existing agent i if and only if

$$\text{CE}_i(\lambda_{n,i}S_n - \pi_{n,i}) \geq \text{CE}_i(\lambda_{n+1,i}S_{n+1} - \pi_{n+1,i}).$$

It follows that

$$\frac{s_n^2}{\Delta_n^2} \geq \frac{s_{n+1}^2}{\Delta_{n+1}^2} \quad (4)$$

Similarly, the candidate is willing to participate in the pool if and only of

$$\text{CE}_{n+1}(X_{n+1}) \geq \text{CE}_{n+1}(\lambda_{n+1,n+1}S_{n+1} - \pi_{n+1,n+1}).$$

It follows that

$$\frac{\sigma_{n+1}^2}{\delta_{n+1}^2} \geq \frac{s_{n+1}^2}{\Delta_{n+1}^2}. \quad (5)$$

Proof of Corollary 1

Since $s_{n+1}^2 = s_n^2 + \sigma_{n+1}^2 + 2\rho(S_n, X_{n+1})s_n\sigma_{n+1}$, rewriting the inequality (4) with respect to $\rho(S_n, X_{n+1})$ gives

$$\rho(S_n, X_{n+1}) \leq \frac{(\Delta_{n+1}^2 - \Delta_n^2)s_n^2 - \Delta_n^2\sigma_{n+1}^2}{2\Delta_n^2s_n\sigma_{n+1}}.$$

In a similar manner, rewriting the inequality (5) with respect to $\rho(S_n, X_{n+1})$ gives

$$\rho(S_n, X_{n+1}) \leq \frac{(\Delta_{n+1}^2 - \delta_{n+1}^2)\sigma_{n+1}^2 - \delta_{n+1}^2s_n^2}{2\delta_{n+1}^2s_n\sigma_{n+1}}.$$

Proof of Corollary 2

Recall that $\Sigma_n := \sum_{i=1}^n \sigma_i$. By the normality assumption and the common pairwise correlation assumption,

$$\rho(S_n, X_{n+1}) = \sum_{i=1}^n \frac{\text{Cov}(X_i, X_{n+1})}{s_n \sigma_{n+1}} = \frac{\rho \Sigma_n}{s_n}.$$

Thus, (4) becomes

$$\sigma_{n+1}^2 + 2\rho \Sigma_n \sigma_{n+1} - (\Delta_{n+1}^2 / \Delta_n^2 - 1) s_n^2 \leq 0,$$

which is equivalent to

$$0 < \sigma_{n+1} \leq -\rho \Sigma_n + \sqrt{(\rho \Sigma_n)^2 + \left(\frac{\Delta_{n+1}^2}{\Delta_n^2} - 1 \right) s_n^2}.$$

Moreover, (5) becomes

$$(\Delta_{n+1}^2 - \delta_{n+1}^2) \sigma_{n+1}^2 - 2\rho \delta_{n+1}^2 \sigma_{n+1} \Sigma_n - \delta_{n+1}^2 s_n^2 \geq 0.$$

which is equivalent to

$$\sigma_{n+1} \geq \delta_{n+1}^2 \frac{\rho \Sigma_n + \sqrt{(\rho \Sigma_n)^2 + (\Delta_{n+1}^2 - \delta_{n+1}^2) s_n^2}}{\Delta_{n+1}^2 - \delta_{n+1}^2} > 0.$$

Combining two inequalities gives the desired bounds for σ_{n+1} .

Proof of Remark 4

It suffices to show that for any given $\delta = (\delta_i)_{1 \leq i \leq n}$ and $\sigma = (\sigma_i)_{1 \leq i \leq n}$, there is no σ_{n+1} such that

$$\max \left\{ \frac{(\Delta_{n+1}^2 - \Delta_n^2) s_n^2 - \Delta_n^2 \sigma_{n+1}^2}{2\Delta_n^2 s_n \sigma_{n+1}}, \frac{(\Delta_{n+1}^2 - \delta_{n+1}^2) \sigma_{n+1}^2 - \delta_{n+1}^2 s_n^2}{2\delta_{n+1}^2 s_n \sigma_{n+1}} \right\} < 1.$$

Two functions intersect when

$$\frac{(\Delta_{n+1}^2 - \Delta_n^2) s_n^2 - \Delta_n^2 \sigma_{n+1}^2}{2\Delta_n^2 s_n \sigma_{n+1}} = \frac{(\Delta_{n+1}^2 - \delta_{n+1}^2) \sigma_{n+1}^2 - \delta_{n+1}^2 s_n^2}{2\delta_{n+1}^2 s_n \sigma_{n+1}},$$

which occurs at

$$\sigma_{n+1}^* = \frac{\delta_{n+1} s_n}{\Delta_n}.$$

Note that

$$\left| \frac{(\Delta_{n+1}^2 - \Delta_n^2) s_n^2 - \Delta_n^2 \sigma_{n+1}^2}{2\Delta_n^2 s_n \sigma_{n+1}} \right|_{\sigma_{n+1} = \sigma_{n+1}^*} = 1.$$

But if $\sigma_{n+1} < \delta_{n+1}s_n/\Delta_n$, then

$$\frac{(\Delta_{n+1}^2 - \Delta_n^2)s_n^2 - \Delta_n^2\sigma_{n+1}^2}{2\Delta_n^2s_n\sigma_{n+1}} > 1,$$

and if $\sigma_{n+1} > \delta_{n+1}s_n/\Delta_n$, we have

$$\frac{(\Delta_{n+1}^2 - \delta_{n+1}^2)\sigma_{n+1}^2 - \delta_{n+1}^2s_n^2}{2\delta_{n+1}^2s_n\sigma_{n+1}} > 1.$$

This implies the desired result.

Proof of Theorem 3

The aggregate risk after the expansion is

$$S_{n+1} = \sum_{j=1}^{n+1} X_j \sim \text{Normal}(m_{n+1}, s_{n+1}^2),$$

where

$$m_{n+1} = \sum_{j=1}^{n+1} \mu_j \quad \text{and} \quad s_{n+1}^2 = s_n^2 + \sigma_{n+1}^2 + 2\rho(S_n, X_{n+1})s_n\sigma_{n+1}.$$

By Theorem 1, the risk allocated to the agent i is

$$\lambda_{n+1,i}S_{n+1} \sim \text{Normal}(\lambda_{n+1,i}m_{n+1}, \lambda_{n+1,i}^2s_{n+1}^2).$$

By the fairness principle,

$$\pi_{n+1,i} = \lambda_{n+1,i}m_{n+1} - \mu_i.$$

The existing agent i will leave the risk pool if

$$\text{CE}_i[X_i] \leq \text{CE}_i[\lambda_{n+1,i}S_{n+1} - \pi_{n+1,i}],$$

which is equivalent to

$$\frac{\sigma_i^2}{\delta_i^2} \leq \frac{s_{n+1}^2}{\Delta_{n+1}^2}. \tag{6}$$

Rewriting (6) with respect to $\rho(S_n, X_{n+1})$ gives

$$\rho(S_n, X_{n+1}) \geq \frac{\Delta_{n+1}^2\sigma_i^2 - \delta_i^2\sigma_{n+1}^2 - \delta_i^2s_n^2}{2\delta_i^2s_n\sigma_{n+1}}.$$

E. Supplementary Materials for Section 4

E.1. Maximum Extra Premium

Thanks to the combined effect of risk diversification and reinsurance, each member is in fact willing to pay an extra premium, on top of the actuarially fair premium. This will make the total reinsurance premium equal to $pr^{(n)} + c^{(n)} = \sum_{i=1}^n (pr_{n,i} + c_i) = pr^{(n)} + \mathbb{E}[(S_n - K)_+]$, where $pr_{n,i}$ is the extra premium allocated to agent i , such that $\sum_{i=1}^n pr_{n,i} = pr$. From a different perspective, the above arguments could yield the maximum reinsurance premium pr_n^* that the group is willing to pay and still achieve a positive aggregate risk reduction. For this, we simply consider pr_n^* as the total reduction that the actuarially fair-priced reinsurance gives to the group. This is the subject of the following statement.

Under the normality assumption and actuarial fairness, the maximum extra premium that the pool is willing to pay is given as

$$pr_n^* = \Delta_n \left(\frac{1}{2} \left(\frac{s_n}{\Delta_n} \right)^2 - l \left(\frac{s_n}{\Delta_n}; \beta \right) \right) > 0, \quad (7)$$

where function $l(x; \beta)$ is given as

$$l(x; \beta) := \log \left(e^{\frac{x^2}{2}} \Phi(\beta - x) + e^{\beta x} (1 - \Phi(\beta)) \right) + x(\phi(\beta) - \beta(1 - \Phi(\beta))).$$

The proof is as follows. Let the extra premium (on top of fair price) paid to the reinsurer by the pool be equal to $pr^{(n)}$. We first calculate the aggregate risk measure without reinsurance:

$$\begin{aligned} \sum_{i=1}^n \rho_i [\text{Without Reinsurance}] &= \sum_{i=1}^n \rho_i [\lambda_{n,i} S_n + \pi_{n,i}] = \sum_{i=1}^n \left(\lambda_{n,i} m_n + \frac{\lambda_{n,i}}{2\Delta_n} s_n^2 + \mu_i - \lambda_{n,i} m_n \right) \\ &= m_n + \frac{s_n^2}{2\Delta_n}. \end{aligned}$$

With the reinsurance, the aggregate risk measure is

$$\sum_{i=1}^n \rho_i [\text{With Reinsurance}] = \sum_{i=1}^n \rho_i [\lambda_{n,i} \bar{S}_n + pr_{n,i} - c_i] = \sum_{i=1}^n \rho_i [\lambda_{n,i} \bar{S}_n] + \pi_{n,i} + pr^{(n)} + c^{(n)}.$$

The first part of the above equation is written as

$$\begin{aligned} \sum_{i=1}^n \text{CE}_i [\lambda_{n,i} \bar{S}_n] &= \sum_{i=1}^n \delta_i \log \left(\left(e^{\frac{1}{\Delta_n} m_n + \frac{1}{2\Delta_n^2} s_n^2} \right) \Phi \left(\beta - \frac{1}{\Delta_n} s_n \right) + e^{\frac{1}{\Delta_n} K} (1 - \Phi(\beta)) \right) \\ &= m_n + \sum_{i=1}^n \delta_i l_n = m_n + \Delta_n l_n, \end{aligned}$$

where

$$l_n := l \left(\frac{s_n}{\Delta_n}; \beta \right) = \log \left(e^{\frac{s_n^2}{2\Delta_n^2}} \Phi \left(\beta - \frac{s_n}{\Delta_n} \right) + e^{\beta \frac{s_n}{\Delta_n}} (1 - \Phi(\beta)) \right).$$

Since for the internal cash transfer it holds that $\sum_{i=1}^n \pi_{n,i} = 0$, the total pool's cash payment to the reinsurance is

$$\begin{aligned} pr^{(n)} + c^{(n)} &= pr^{(n)} + \mathbb{E}[(S_n - K)_+] = pr^{(n)} + (\mathbb{E}[S_n | S_n \geq K] - K)\mathbb{P}[S_n \geq K] \\ &= pr^{(n)} + \left(m_n + \frac{s_n \phi(\beta)}{1 - \Phi(\beta)} - K \right) (1 - \Phi(\beta)) \\ &= pr^{(n)} + m_n(1 - \Phi(\beta)) + s_n \phi(\beta) - (m_n + \beta s_n)(1 - \Phi(\beta)) \\ &= pr^{(n)} + s_n(\phi(\beta) - \beta(1 - \Phi(\beta))). \end{aligned}$$

Therefore,

$$\sum_{i=1}^n \text{CE}_i[\text{With Reinsurance}] = m_n + \Delta_n l_n + pr^{(n)} + s_n(\phi(\beta) - \beta(1 - \Phi(\beta))).$$

The maximum acceptable extra premium pr^* is the one that makes aggregate risk measures equal:

$$\sum_{i=1}^n \rho_i[\text{Without Reinsurance}] = \sum_{i=1}^n \rho_i[\text{With Reinsurance}],$$

which gives (7).

E.2. Proofs of Section 4

Proof of Theorem 4

Define

$$g(x; \beta) = \frac{x^2}{2} - l(x; \beta).$$

It suffices to show that $g(x; \beta) \geq 0$ for all $x, \beta > 0$. Note that $g(0; \beta) = 0$ for all β . Fix $\beta > 0$. Then,

$$\frac{d}{dx} g(x; \beta) = x - \frac{x e^{x^2/2} \Phi(\beta - x) - e^{x^2/2} \phi(\beta - x) + \beta e^{x\beta} \Phi(-\beta)}{e^{x^2/2} \Phi(\beta - x) + e^{x\beta} \Phi(-\beta)} - \phi(\beta) + \beta \Phi(-\beta).$$

Thus, $dg(x; \beta)/dx \geq 0$ if and only if

$$\begin{aligned} & x e^{x\beta} \Phi(-\beta) + e^{x^2/2} \phi(\beta - x) - \beta e^{x\beta} \Phi(-\beta) - \phi(\beta) e^{x^2/2} \Phi(\beta - x) \\ & - \phi(\beta) e^{x\beta} \Phi(-\beta) + \beta \Phi(-\beta) e^{x^2/2} \Phi(\beta - x) + \beta \Phi(-\beta) e^{x\beta} \Phi(-\beta) \geq 0. \end{aligned} \tag{8}$$

Replacing $\phi(z)$ with $e^{-z^2/2}/\sqrt{2\pi}$ and multiplying $e^{-x\beta+\beta^2/2}$ to (8) makes it equivalent to

$$\begin{aligned} & x e^{\beta^2/2} \Phi(-\beta) + 1/\sqrt{2\pi} + \beta e^{(x-\beta)^2/2} \Phi(-\beta) \Phi(\beta - x) + \beta \Phi^2(-\beta) e^{\beta^2/2} \\ & \geq \beta e^{\beta^2/2} \Phi(-\beta) + e^{x^2/2-x\beta} \Phi(\beta - x)/\sqrt{2\pi} + \Phi(-\beta)/\sqrt{2\pi}. \end{aligned} \tag{9}$$

Assume for now that $x > \beta$. In the view of (9), we directly get that

$$\begin{aligned} & x e^{\beta^2/2} \Phi(-\beta) > \beta e^{\beta^2/2} \Phi(-\beta) \\ & 0.5/\sqrt{2\pi} > \Phi(-\beta)/\sqrt{2\pi}. \end{aligned}$$

On the other hand, we it also holds that

$$0.5/\sqrt{2\pi} + \beta e^{(x-\beta)^2/2} \Phi(-\beta) \Phi(\beta-x) > e^{x^2/2-x\beta} \Phi(\beta-x)/\sqrt{2\pi}.$$

Indeed, the above inequality can be re-written as

$$\frac{0.5}{\Phi(\beta-x)e^{x^2/2-x\beta}} + \sqrt{2\pi}\beta e^{\beta^2/2} \Phi(-\beta) > 1.$$

Note that

$$\frac{d}{dx} \Phi(\beta-x)e^{x^2/2-x\beta} = e^{x^2/2-x\beta} \phi(\beta-x) \left(\frac{(x-\beta)\Phi(\beta-x)}{\phi(\beta-x)} - 1 \right)$$

By the Mill's ratio,

$$\frac{(x-\beta)\Phi(\beta-x)}{\phi(\beta-x)} = \frac{(x-\beta)(1-\Phi(x-\beta))}{\phi(x-\beta)} < 1.$$

Since $d\Phi(\beta-x)e^{x^2/2-x\beta}/dx < 0$ for all $x > \beta$,

$$\frac{0.5}{\Phi(\beta-x)e^{x^2/2-x\beta}} + \sqrt{2\pi}\beta e^{\beta^2/2} \Phi(-\beta) > e^{\beta^2/2} + \sqrt{2\pi}\beta e^{\beta^2/2} \Phi(-\beta) > 1.$$

Hence, we get the desired inequality when $x > \beta$.

Assume now that $0 < x < \beta$. Rewriting (9),

$$(x-\beta\Phi(\beta))e^{\beta^2/2}\Phi(-\beta) + \frac{\Phi(\beta)}{\sqrt{2\pi}} + \frac{\Phi(\beta-x)e^{x^2/2-x\beta}}{\sqrt{2\pi}}(\beta\sqrt{2\pi}e^{\beta^2/2}\Phi(-\beta)-1) \geq 0. \quad (10)$$

Note

$$\frac{d}{dx}(x-\beta\Phi(\beta))e^{\beta^2/2}\Phi(-\beta) = e^{\beta^2/2}\Phi(-\beta) > 0$$

By the Mill's ratio,

$$\beta\sqrt{2\pi}e^{\beta^2/2}\Phi(-\beta) = \beta \frac{1-\Phi(\beta)}{\frac{1}{\sqrt{2\pi}}e^{-\beta^2/2}} = \beta \frac{1-\Phi(\beta)}{\phi(\beta)} < 1.$$

Thus,

$$\begin{aligned} & \frac{d}{dx} \Phi(\beta-x)e^{x^2/2-x\beta}(\beta\sqrt{2\pi}e^{\beta^2/2}\Phi(-\beta)-1) \\ &= e^{x^2/2-x\beta}(\beta\sqrt{2\pi}e^{\beta^2/2}\Phi(-\beta)-1)(-\phi(\beta-x)+\phi(\beta-x)(x-\beta)) > 0. \end{aligned}$$

It follows that LHS of (10) is non-decreasing with respect to x . Then,

$$\begin{aligned} & (x-\beta\Phi(\beta))e^{\beta^2/2}\Phi(-\beta) + \frac{\Phi(\beta)}{\sqrt{2\pi}} + \frac{\Phi(\beta-x)e^{x^2/2-x\beta}}{\sqrt{2\pi}}(\beta\sqrt{2\pi}e^{\beta^2/2}\Phi(-\beta)-1) \\ & \geq \lim_{x \rightarrow 0} (x-\beta\Phi(\beta))e^{\beta^2/2}\Phi(-\beta) + \frac{\Phi(\beta)}{\sqrt{2\pi}} + \frac{\Phi(\beta-x)e^{x^2/2-x\beta}}{\sqrt{2\pi}}(\beta\sqrt{2\pi}e^{\beta^2/2}\Phi(-\beta)-1) \\ & = \Phi(\beta)\Phi(-\beta)e^{\beta^2/2}(-\beta+\beta) \\ & = 0. \end{aligned}$$

Since $dg(x; \beta)/dx$ is continuous and non-negative for $0 < x < \beta$ and $x > \beta$, it is also non-negative when $x = \beta$. It follows that $dg(x; \beta)/dx \geq 0$ for all $x, \beta > 0$. For all $x, \beta > 0$ and hence

$$g(x; \beta) \geq g(0; \beta) = 0.$$

Proof of Theorem 6

For the existing members, it must hold that

$$\text{CE}_i[\text{n agents with reinsurance}] - \text{CE}_i[\text{n+1 agents with Reinsurance}] \geq 0,$$

i.e.,

$$f_{e,i} = \text{CE}_i [\lambda_{n,i} \bar{S}_n - \pi_{n,i} - p_{n,i}] - \text{CE}_i [\lambda_{n+1,i} \bar{S}_{n+1} - \pi_{n+1,i} - p_{n+1,i}] \geq 0.$$

That is,

$$f_{e,i} = \delta_i l_n - \lambda_{n,i} s_n (\phi(\beta) - \beta(1 - \Phi(\beta))) + \mu_i - \delta_i l_{n+1} + \lambda_{n+1,i} s_{n+1} (\phi(\beta) - \beta(1 - \Phi(\beta))) - \mu_i \geq 0.$$

Thus,

$$f_{e,i} = \delta_i (l_n - l_{n+1}) - (\lambda_{n,i} s_n - \lambda_{n+1,i} s_{n+1}) (\phi(\beta) - \beta(1 - \Phi(\beta))) \geq 0.$$

Dividing by δ_i gives the desired result. On the other hand, for the candidate to join, it must hold that

$$f_c = \text{CE}_{n+1} [X_{n+1}] - \text{CE}_{n+1} [\lambda_{n+1,n+1} \bar{S}_{n+1} - \pi_{n+1,n+1} - p_{n+1,n+1}] \geq 0.$$

As above,

$$f_c = \frac{1}{2\delta_{n+1}} \sigma_{n+1}^2 - \delta_{n+1} l_{n+1} + \lambda_{n+1,n+1} s_{n+1} (\phi(\beta) - \beta(1 - \Phi(\beta))).$$

Proof of Theorem 7

To prove Theorem 7, we need to prove the following statement.

Let $M : \mathbb{R} \mapsto \mathbb{R}$ be defined by $M(\beta) := \beta + \frac{1}{\beta}$. The function $l(x; \beta)$ monotonically increases with respect to $x > 0$, for all $\beta > 0$, when $x \geq M(\beta)$.

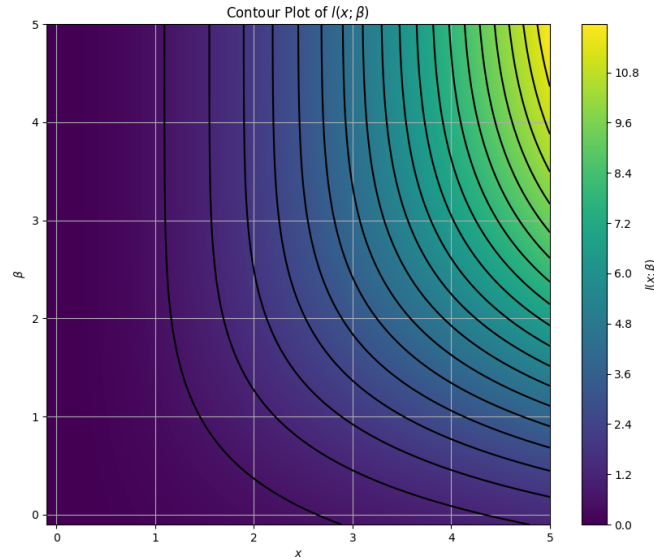
It suffices to show that $l(x; \beta) \geq l(y; \beta)$ for any $x \geq y > M(\beta)$ and $\beta > 0$.

$$l(x; \beta) - l(y; \beta) = \log \left(\frac{e^{\frac{x^2}{2}} \Phi(\beta - x) + e^{\beta x} (1 - \Phi(\beta))}{e^{\frac{y^2}{2}} \Phi(\beta - y) + e^{\beta y} (1 - \Phi(\beta))} \right) + (x - y) (\phi(\beta) - \beta(1 - \Phi(\beta))). \quad (11)$$

Note that

$$\lim_{\beta \rightarrow \infty} \phi(\beta) - \beta(1 - \Phi(\beta)) = 0 \quad \text{and} \quad \frac{d}{d\beta} (\phi(\beta) - \beta(1 - \Phi(\beta))) = \Phi(\beta) - 1 \leq 0.$$

Figure 8 Contour plot of $l(x; \beta)$



Thus, $\phi(\beta) - \beta(1 - \Phi(\beta)) \geq 0$ for all β meaning that the second term in (11) is non-negative. Moreover, we readily get that

$$(e^{\beta x} - e^{\beta y})(1 - \Phi(\beta)) \geq 0,$$

which means that it is only left to show that $e^{\frac{x^2}{2}} \Phi(\beta - x)$ is monotone for $x \geq M(\beta)$ and $\beta > 0$. For this, we calculate

$$\frac{d}{dx} \left(e^{\frac{x^2}{2}} \Phi(\beta - x) \right) = e^{\frac{x^2}{2}} (x\Phi(\beta - x) - \phi(\beta - x)),$$

which is non-negative if

$$\frac{\Phi(\beta - x)}{\phi(\beta - x)} \geq \frac{1}{x}.$$

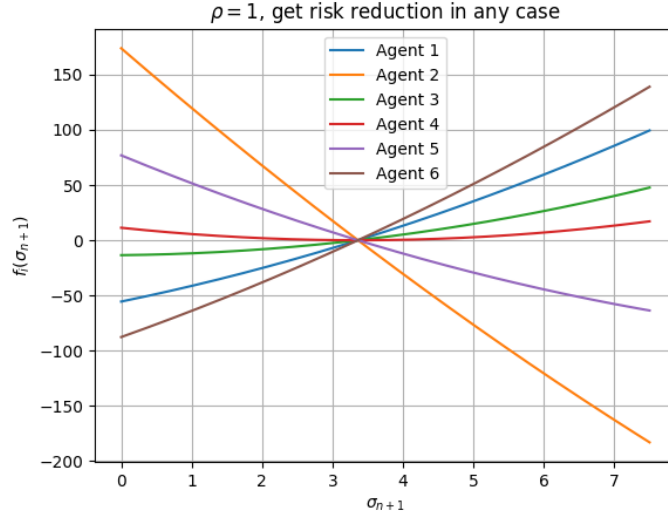
The lower bound of the Mill's ratio implies that

$$\frac{\Phi(\beta - x)}{\phi(\beta - x)} - \frac{1}{x} \geq \frac{x - \beta}{(x - \beta)^2 + 1} - \frac{1}{x} = \frac{\beta x - \beta^2 - 1}{x((x - \beta)^2 + 1)} \geq 0$$

which indeed holds for any x such that

$$x \geq \beta + \frac{1}{\beta}.$$

Note that $M(\beta)$ is not a sharp boundary above which the function l becomes monotonic. In the statement, it is proved analytically that $l(x; \beta)$ monotonically increases when $x \geq M(\beta)$ for any given β . However, numerically, we could verify that $l(x; \beta)$ monotonically increases when $x > 0$ for any given β . In fact, the contour plot of $l(x; \beta)$ is exhibited in Figure 8. Then, Theorem 7 follows from Theorem 6 and the above statement.

Figure 9 Counter-example of Corollary 3 when heterogeneous risk tolerances are assumed.

Note: The parameters are $\rho = 1$, $\sigma_6 \approx (2.7, 12.6, 9, 20, 19.6, 0.2)$, $\delta_7 \approx (13.5, 7.1, 15.7, 19.2, 13, 7.2, 14.4)$.

F. Supplementary Materials for Section 5

F.1. Counter-example: Corollary 3 for heterogeneous risk tolerances

Recall that the i -th agent gets benefited by the entrance if and only if $f_i(\sigma_{n+1}) > 0$, where $f_i(\sigma_{n+1})$ for $i = 1, 2, \dots, 6$ are defined in (13) from the main text. In Figure 9, We see that the entrance of the candidate always leads to risk reduction for the fourth agent since $f_4(\sigma_{n+1}) > 0$ for all σ_{n+1} .

F.2. Proofs of Section 5

Proof of Theorem 9

Note that $\rho(Y_i^{(n)}) \geq \rho(Y_i^{(n+1)})$ is equivalent to $\rho(X_i) - \rho(Y_i^{(n)}) \leq \rho(X_i) - \rho(Y_i^{(n+1)})$ and the result follows directly from (17).

Proof of Theorem 10

For the first item in the view to Table 1, we need to show that when σ_i is sufficiently low for some $i = 1, \dots, n$, then $b_i^2 \leq c_i$. Indeed, we calculate that $b_i^2 \leq c_i$ is equivalent to

$$2 \left(\frac{\lambda_{n,i} - \lambda_{n+1,i}}{\lambda_{n+1,i}^2} - \frac{\lambda_{n,i}^2 - \lambda_{n+1,i}^2}{\lambda_{n+1,i}^2} \right) \sigma_i^2 + 2\rho\sigma_i \Sigma_{-i}^{(n)} \left(\frac{\lambda_{n,i} - \lambda_{n+1,i}}{\lambda_{n+1,i}^2} - \frac{\lambda_{n,i}^2 - \lambda_{n+1,i}^2}{\lambda_{n+1,i}^2} \right) - \frac{\lambda_{n,i}^2 - \lambda_{n+1,i}^2}{\lambda_{n+1,i}^2} s_{-i}^2 \leq 0.$$

Since $\lambda_{n,i} > \lambda_{n+1,i}$, the above holds when σ_i is sufficiently low.

As for the second item, we need to show for large σ_i and ρ we have $0 \geq c_i < b_i^2$ and also $b_i > 0$. Indeed, the latter holds exactly when $\sigma_i > \lambda_{n+1,i} \Sigma_{-i}^{(n)} / (1 - \lambda_{n+1,i})$. For the former is equivalent to $f_i(b_i) \leq 0$. Note

that $f_i(b_i)$ is a quadratic function of σ_i with the coefficient of σ_i^2 equal to

$$2 \frac{\lambda_{n,i} - \lambda_{n+1,i}}{\lambda_{n+1,i}^2} - \frac{\lambda_{n,i}^2 - \lambda_{n+1,i}^2}{\lambda_{n+1,i}^2} - \rho^2 \frac{(1 - \lambda_{n+1,i})^2}{\lambda_{n+1,i}^2},$$

which is negative if and only if

$$\rho^2 > \frac{(\lambda_{n,i} - \lambda_{n+1,i})(2 - (\lambda_{n,i} + \lambda_{n+1,i}))}{(1 - \lambda_{n+1,i})^2}.$$

Hence, when both ρ and σ_i are sufficiently large, function $f_i(x)$ is positive at zero, becomes negative for a positive x^* and then increases. This behavior is consistent with the statement.

Finally, for the third item it is enough to show that $f_i(b_i) \geq 0$. $f_i(b_i)$ is a quadratic function of σ_i , the coefficient of σ_i^2 becomes positive when

$$\rho^2 < \frac{(\lambda_{n,i} - \lambda_{n+1,i})(2 - (\lambda_{n,i} + \lambda_{n+1,i}))}{(1 - \lambda_{n+1,i})^2}.$$

The above means that for sufficiently large σ_i and when ρ satisfies the above inequality, function $f_i(b_i)$ is always above zero and hence agent i always accept the candidate.

Proof of Corollary 3

Note that

$$\lim_{\rho \rightarrow 1} f_i(\sigma_{n+1} | \sigma) = \sigma_{n+1}^2 + 2(\Sigma_n - (n+1)\sigma_i)\sigma_{n+1} + \frac{2n+2}{n} \left(\sigma_i^2 + \sigma_i \sum_{j \neq i}^n \sigma_j \right) - \frac{2n+1}{n^2} \left(\sum_{j=1}^n \sigma_j^2 + 2 \sum_{j \neq k}^n \sigma_j \sigma_k \right).$$

The minimum is achieved at

$$\frac{\partial}{\partial \sigma_{n+1}} \lim_{\rho \rightarrow 1} f_i(\sigma_{n+1} | \sigma) = 2\sigma_{n+1} + 2(\Sigma_n - (n+1)\sigma_i) = 0.$$

Thus, $\sigma_{n+1}^* = (n+1)\sigma_i - \Sigma_n$. First, suppose that $(n+1)\sigma_i - \Sigma_n < 0$. Since the minimum is achieved when $\sigma_{n+1} < 0$, it suffices to prove that $f_i(\sigma_{n+1} = 0 | \sigma) < 0$. Define $\Sigma_{n-i} := \sum_{j \neq i}^n \sigma_j$. Then,

$$\begin{aligned} \lim_{\rho \rightarrow 1} f_i(\sigma_{n+1} = 0 | \sigma) &= \frac{2n+2}{n} \left(\sigma_i^2 + \sigma_i \sum_{j \neq i}^n \sigma_j \right) - \frac{2n+1}{n^2} \left(\sum_{j=1}^n \sigma_j^2 + 2 \sum_{j \neq k}^n \sigma_j \sigma_k \right) \\ &= \frac{1}{n^2} ((2n^2 - 1)\sigma_i^2 + 2(n^2 - n - 1)\sigma_i \Sigma_{n-i} - (2n+1)\Sigma_{n-i}^2) \\ &= (\sigma_i + \Sigma_{n-i})((2n^2 - 1)\sigma_i - (2n+1)\Sigma_{n-i}) \\ &< 0, \end{aligned}$$

where the last inequality comes from the imposed assumption.

Second, suppose that $(n+1)\sigma_i - \Sigma_n > 0$. Since the minimum is achieved when $\sigma_{n+1} > 0$, we have to prove that $f_i(\sigma_{n+1} = \sigma_{n+1}^* | \sigma) < 0$. Note the minimum value is

$$\begin{aligned} \lim_{\rho \rightarrow 1} f_i(\sigma_{n+1} | \sigma) |_{\sigma_{n+1} = \sigma_{n+1}^*} &= -((n+1)\sigma_i - \Sigma_n)^2 + \frac{2n+2}{n} \left(\sigma_i^2 + \sigma_i \sum_{j \neq i}^n \sigma_j \right) - \frac{2n+1}{n^2} \left(\sum_{j=1}^n \sigma_j^2 + 2 \sum_{j \neq k}^n \sigma_j \sigma_k \right) \\ &= -\frac{1}{n^2} ((n^2-1)^2 \sigma_i^2 - 2(n-1)(n+1)^2 \sigma_i \Sigma_{n-i} + (n+1)^2 \Sigma_{n-i}^2) \\ &= -\frac{1}{n^2} (1+n)^2 (\sigma_i - n\sigma_i + \Sigma_{n-i})^2 \\ &< 0, \end{aligned}$$

which completes the proof.

Proof of Theorem 11

The first item follows directly since $f_i(\sigma_{n+1} | \sigma)$ is a quadratic function with a positive constant at the quadratic term. Thus, it is positive for sufficiently large σ_{n+1} for all i in any case.

For the second item, if σ_{n+1} is sufficiently low, the strong consensus condition is determined by the existing member with the lower of f_i . For this, we consider two existing members i, j and compare the limiting behavior of f_i and f_j when σ_{n+1} goes to zero. We calculate that

$$\lim_{\sigma_{n+1} \rightarrow 0} [f_i(\sigma_{n+1}) - f_j(\sigma_{n+1})] = 2 \frac{\delta_{n+1} \Delta_{n+1}}{\Delta_n} \left(\frac{\sigma_i}{\delta_i} (\sigma_i - \rho \Sigma_{-i}) - \frac{\sigma_j}{\delta_j} (\sigma_j - \rho \Sigma_{-j}) \right),$$

where we have used the fact that $\lambda_{n,i}^2 / \lambda_{n+1,i}^2$ does not depend on i . The above is negative if and only if

$$\frac{\sigma_i^2}{\delta_i} - \frac{\sigma_j^2}{\delta_j} + \rho \frac{\sigma_i}{\delta_i} \frac{\sigma_j}{\delta_j} (\delta_j - \delta_i) + \rho \Sigma_{-i,j} \left(\frac{\sigma_i}{\delta_i} - \frac{\sigma_j}{\delta_j} \right) < 0,$$

where $\Sigma_{-i,j} := \sum_{k \neq i,j}^n \sigma_k$. Thus, we may conclude that the existing member with sufficiently low risk and sufficiently high risk tolerance (that is sufficiently low deviation-tolerance ratio) determines the strong consensus condition.

For the third item, we analogously examine the sign of the difference $f_i(\sigma_{n+1}) - f_j(\sigma_{n+1})$ when σ_{n+1} is sufficiently large. After simple calculations we get that this difference is negative if

$$\Delta_{n+1} \left(\frac{\sigma_j}{\delta_j} - \frac{\sigma_i}{\delta_i} \right) < 0.$$

Thus for sufficiently large σ_{n+1} , the existing member with the higher deviation-tolerance ratio.

Proof of Theorem 12

Imposing common variances $\sigma_i = \sigma$ and common risk tolerances $\delta_i = \delta$ for each $i = 1, 2, \dots, n$, (19) becomes

$$\sigma_{n+1}^2 - 2\rho\sigma\sigma_{n+1} + \frac{2(n+1)(1+(n-1)\rho)}{n}\sigma^2 - \frac{(2n+1)(1+(n-1)\rho)}{n}\sigma^2 \geq 0,$$

which reduces to

$$\sigma_{n+1}^2 - 2\rho\sigma\sigma_{n+1} + \frac{(1 + (n-1)\rho)}{n}\sigma^2 \geq 0,$$

This is quadratic convex function of σ_{n+1} . Then, note that

$$(\rho\sigma)^2 - \frac{(1 + (n-1)\rho)}{n}\sigma^2 < 0,$$

if $(\rho - 1)(n\rho + 1) < 0$. This condition holds for $-1/n < \rho < 1$, i.e. when ρ is positive or not too negative, which in turn implies that there is no real root for $\rho > 0$. Thus all existing members get benefited by the candidate's entrance.

On the other hand, imposing only common variances simplifies function f_i as below:

$$\begin{aligned} f_i(\sigma_{n+1} | \rho, \sigma, \delta) &= \sigma_{n+1}^2 + 2\frac{\sigma\rho}{\lambda_{n+1,i}}(n\lambda_{n+1,i} - 1)\sigma_{n+1} \\ &\quad + 2\left(\frac{\lambda_{n,i} - \lambda_{n+1,i}}{\lambda_{n+1,i}^2}\right)\sigma^2(1 + \rho(n-1)) - \frac{\lambda_{n,i}^2 - \lambda_{n+1,i}^2}{\lambda_{n+1,i}^2}\sigma^2n(1 + \rho(n-1)). \end{aligned}$$

The necessary condition for $A = \mathbb{R}_+$ for all δ_{n+1} is

$$f_i(0) = \sigma^2(1 + \rho(n-1))\frac{\lambda_{n,i} - \lambda_{n+1,i}}{\lambda_{n+1,i}^2}[2 - n(\lambda_{n,i} + \lambda_{n+1,i})] > 0,$$

which becomes equivalent to $\lambda_{n,i} + \lambda_{n+1,i} < (2/n)$ for $n > 1$. We then need $b_i \leq 0$ or $f_i(b_i) = f_i(0) - b_i^2 \geq 0$ for each $i = 1, 2, \dots, n$. Since

$$b_i = -\frac{\sigma\rho}{\lambda_{n+1,i}}(n\lambda_{n+1,i} - 1),$$

the former means that $1/n < \lambda_{n+1,i}$, which cannot hold for all i . Hence, there is at least one existing member for which $f_i(b_i) = f_i(0) - b_i^2 \geq 0$. However, taking $\delta_{n+1} \rightarrow 0$ gives

$$f_i(b_i) \rightarrow -\frac{\sigma^2\rho^2}{\lambda_{n,i}^2}(n\lambda_{n,i} - 1)^2 < 0, \quad \text{if } \lambda_{n,i} \neq \frac{1}{n},$$

which proves the desired result.

As for the last part of the statement the following example (pictured in Figure 10) provides the mentioned possibility. Consider $n = 6$, $\rho \approx 0.96$, common $\sigma = 1$, and $\delta_{n+1} \approx (6.4, 7.1, 6.2, 6.4, 6.8, 6.8, 8.4)$.

Proof of Theorem 13

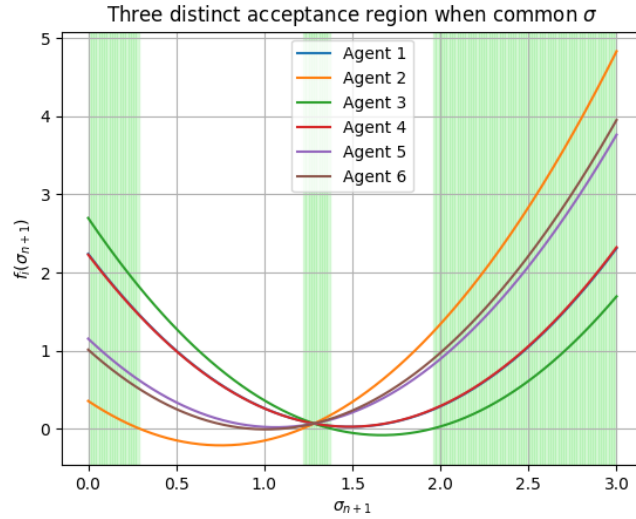
When existing members' risk tolerances are fixed equal to δ , (18) becomes

$$f_i(\sigma_{n+1} | \sigma) := \sigma_{n+1}^2 + 2\rho(\Sigma_n - (n+1)\sigma_i)\sigma_{n+1} + \frac{2n+2}{n}\left(\sigma_i^2 + \rho\sigma_i \sum_{j \neq i}^n \sigma_j\right) - \frac{2n+1}{n^2}\left(\sum_{j=1}^n \sigma_j^2 + 2\rho \sum_{j \neq k}^n \sigma_j\sigma_k\right),$$

where $\sigma = (\sigma_i)_{i=1, \dots, n}$. We prove the argument by proving an example of $n = 2$ and assuming that $\delta_1 = \delta_2 = \delta_3$. We directly calculate that

$$\begin{aligned} f_1(x | \rho, \sigma) &= x^2 + 2\rho(\sigma_2 - 2\sigma_1)x + 3(\sigma_1^2 + \rho\sigma_1\sigma_2) - \frac{5}{4}(\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2) \\ f_2(x | \rho, \sigma) &= x^2 + 2\rho(\sigma_1 - 2\sigma_2)x + 3(\sigma_2^2 + \rho\sigma_1\sigma_2) - \frac{5}{4}(\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2). \end{aligned}$$

Figure 10 Three distinct acceptance region when common σ . Green shaded regions consist the acceptance region.



Assuming without loss of generality $\sigma_1 < \sigma_2$. This directly implies that $b_1 < b_2$. We will show that there is a combination of inputs σ_1, σ_2 and ρ for which

$$A = [0, b_1 - \sqrt{c_1}] \cup [b_1 + \sqrt{c_1}, b_2 - \sqrt{c_2}] \cup [b_2 - \sqrt{c_2}, \infty).$$

Based on the Table 1, we get that the following form of A occurs when $0 < b_1$, $0 < f_1(0)$, $f_2(b_2) < 0$ and $b_1 + \sqrt{c_1} < b_2 - \sqrt{c_2}$. The first requirement is equivalent with the inequality $\sigma_2 < 2\sigma_1$. On the other hand, if $f_1(0) > 0$ then we also have that $f_2(0) > 0$ (since $\sigma_1 < \sigma_2$). The former is equivalent to

$$7\sigma_1^2 - 5\sigma_2^2 + 2\rho\sigma_1\sigma_2 > 0.$$

The above holds for all $\sigma_1 > \sigma_2(\sqrt{\rho^2 + 35} - \rho)/7$. Note that this inequality is consistent with the $\sigma_1 < \sigma_2 < 2\sigma_1$. The inequality $f_2(b_2) < 0$ is equivalent to

$$\sigma_2^2 \frac{7 - 16\rho^2}{4} - \sigma_1^2 \frac{5 + 4\rho^2}{4} + \sigma_1\sigma_2\rho(4\rho - 3) \leq 0.$$

The above is decreasing in ρ and has a unique root in $\hat{\rho} \in [0, 1]$, at which and above the inequality holds. Finally, $b_1 + \sqrt{c_1} < b_2 - \sqrt{c_2}$ is equivalent to

$$\sqrt{-f_1(b_1)} + \sqrt{-f_2(b_2)} < 3\rho(\sigma_2 - \sigma_1).$$

G. Supplementary Materials for Section 6

G.1. Analytic expression for SC with reinsurance EP

We give the analytic formulas for the inequality's components $\mathbb{E}[e^{\bar{S}_n/\Delta_n}]$, $\bar{\mathbb{E}}_n[X_i]$, $\bar{\mathbb{E}}_n[S_n]$, and $\mathbb{E}[(S_n - K_n)_+]$.

1. We have known formula for the moment generating function of the truncated normal random variable. That gives

$$\begin{aligned}\mathbb{E}[e^{\bar{S}_n/\Delta_n}] &= \mathbb{E}[e^{S_n/\Delta_n} | S_n < K_n] \mathbb{P}[S_n < K_n] + \mathbb{E}[e^{K_n/\Delta_n} | S_n \geq K_n] \mathbb{P}[S_n \geq K_n] \\ &= \left(e^{\frac{1}{\Delta_n} m_n + \frac{1}{2\Delta_n^2} s_n^2} \right) \Phi\left(\beta - \frac{1}{\Delta_n} s_n\right) + e^{(m_n + s_n \beta)/\Delta_n} (1 - \Phi(\beta)).\end{aligned}$$

2. We already have the denominator, $\mathbb{E}[e^{\bar{S}_n/\Delta_n}]$, so we only solve for the numerator. Note that

$$\mathbb{E}[X_i e^{\bar{S}_n/\Delta_n}] = \mathbb{E}[X_i e^{\bar{S}_n/\Delta_n} \mathbb{1}_{\{S_n \leq K_n\}}] + \mathbb{E}[X_i e^{\bar{S}_n/\Delta_n} \mathbb{1}_{\{S_n > K_n\}}].$$

Define $A_n := K_n - \sum_{j \neq i}^n X_j \sim \text{Normal}(\mu_a, \sigma_a^2)$. Note that

$$\mathbb{E}[X_i e^{\bar{S}_n/\Delta_n} \mathbb{1}_{\{S_n > K_n\}}] = e^{K_n/\Delta_n} \mathbb{E}[X_i \mathbb{1}_{\{S_n > K_n\}}] = e^{K_n/\Delta_n} \mathbb{E}[X_i \mathbb{1}_{\{X_i > A_n\}}].$$

By the law of total expectation,

$$\mathbb{E}[X_i \mathbb{1}_{\{X_i \geq A_n\}}] = \mathbb{E}[\mathbb{E}[X_i \mathbb{1}_{\{X_i \geq A_n\}} | X_i]].$$

Evaluating the expectation,

$$\mathbb{E}[X_i \mathbb{1}_{\{X_i \geq A_n\}} | X_i] = \int_{-\infty}^{\infty} x_i \mathbb{1}_{\{x_i \geq A_n\}} f_{X_i}(x_i) dx_i = \int_{A_n}^{\infty} x_i f_{X_i}(x_i) dx_i.$$

Then,

$$\begin{aligned}\int_{A_n}^{\infty} x_i f_{X_i}(x_i) dx_i &= -\frac{\sigma_i}{\sqrt{2\pi}} \int_{A_n}^{\infty} -\frac{x_i - \mu_i}{\sigma_i^2} e^{-\frac{(x_i - \mu_i)^2}{2\sigma_i^2}} dx_i + \frac{\mu_i}{\sqrt{2\pi}\sigma_i} \int_{A_n}^{\infty} e^{-\frac{(x_i - \mu_i)^2}{2\sigma_i^2}} dx_i \\ &= \frac{\sigma_i}{\sqrt{2\pi}} e^{-\frac{(A_n - \mu_i)^2}{2\sigma_i^2}} + \mu_i \mathbb{P}[X_i \geq A_n] = \frac{\sigma_i}{\sqrt{2\pi}} e^{-\frac{(A_n - \mu_i)^2}{2\sigma_i^2}} + \mu_i \mathbb{P}[S_n \geq K_n] \\ &= \frac{\sigma_i}{\sqrt{2\pi}} e^{-\frac{(A_n - \mu_i)^2}{2\sigma_i^2}} + \mu_i (1 - \Phi(\beta))\end{aligned}$$

Thus,

$$\mathbb{E}[\mathbb{E}[X_i \mathbb{1}_{\{X_i \geq A_n\}} | X_i]] = \mathbb{E}\left[\frac{\sigma_i}{\sqrt{2\pi}} e^{-\frac{(A_n - \mu_i)^2}{2\sigma_i^2}}\right] + \mu_i (1 - \Phi(\beta)).$$

For the first term,

$$\begin{aligned}\mathbb{E}\left[e^{-\frac{(A_n - \mu_i)^2}{2\sigma_i^2}}\right] &= \int_{-\infty}^{\infty} e^{-\frac{(a_n - \mu_i)^2}{2\sigma_i^2}} \frac{1}{\sqrt{2\pi}\sigma_a} e^{-\frac{(a_n - \mu_a)^2}{2\sigma_a^2}} da_n \\ &= \frac{e^{-\frac{1}{2}\left(\frac{\mu_i^2 \sigma_a^2 + \sigma_i^2 \mu_a^2}{\sigma_i^2 \sigma_a^2} - \frac{(\mu_i^2 \sigma_a^2 + \sigma_i^2 \mu_a^2)^2}{\sigma_a^2 \sigma_i^2 (\sigma_a^2 + \sigma_i^2)}\right)}}{\sqrt{2\pi}\sigma_a} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\frac{\left(a_n - \frac{(\mu_i^2 \sigma_a^2 + \sigma_i^2 \mu_a^2)}{\sigma_a^2 \sigma_i^2 (\sigma_a^2 + \sigma_i^2)}\right)^2}{\frac{\sigma_i^2 \sigma_a^2}{\sigma_a^2 + \sigma_i^2}}} da_n\end{aligned}$$

$$\begin{aligned}
&= \frac{e^{-\frac{1}{2} \left(\frac{\mu_i^2 \sigma_a^2 + \sigma_i^2 \mu_a^2}{\sigma_i^2 \sigma_a^2} - \frac{(\mu_i^2 \sigma_a^2 + \sigma_i^2 \mu_a^2)^2}{\sigma_a^2 \sigma_i^2 (\sigma_a^2 + \sigma_i^2)} \right)}}{\sqrt{2\pi} \sigma_a} \sqrt{2\pi \frac{\sigma_i^2 \sigma_a^2}{\sigma_a^2 + \sigma_i^2}} \\
&= \frac{\sigma_i e^{-\frac{1}{2} \left(\frac{\mu_i^2 \sigma_a^2 + \sigma_i^2 \mu_a^2}{\sigma_i^2 \sigma_a^2} - \frac{(\mu_i^2 \sigma_a^2 + \sigma_i^2 \mu_a^2)^2}{\sigma_a^2 \sigma_i^2 (\sigma_a^2 + \sigma_i^2)} \right)}}{\sqrt{\sigma_a^2 + \sigma_i^2}}
\end{aligned}$$

Therefore,

$$\mathbb{E}[X_i \mathbb{1}_{\{X_i \geq A_n\}}] = \frac{\sigma_i^2 e^{-\frac{1}{2} \left(\frac{\mu_i^2 \sigma_a^2 + \sigma_i^2 \mu_a^2}{\sigma_i^2 \sigma_a^2} - \frac{(\mu_i^2 \sigma_a^2 + \sigma_i^2 \mu_a^2)^2}{\sigma_a^2 \sigma_i^2 (\sigma_a^2 + \sigma_i^2)} \right)}}{\sqrt{2\pi(\sigma_a^2 + \sigma_i^2)}} + \mu_i(1 - \Phi(\beta)).$$

Then,

$$\begin{aligned}
\mathbb{E}[X_i e^{\bar{S}_n / \Delta_n} \mathbb{1}_{\{S_n \leq K_n\}}] &= \mathbb{E}[X_i e^{S_n / \Delta_n} | S_n < K_n] \mathbb{P}[S_n < K_n] \\
&= \mathbb{E}[X_i e^{S_n / \Delta_n} | S_n < K_n] \Phi(\beta)
\end{aligned}$$

Let $\mathbf{X} = (X_1, X_2, \dots, X_n) \sim \text{Normal}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Then, $\mathbf{t}^\top \mathbf{X} \sim \text{Normal}(\mathbf{t}^\top \boldsymbol{\mu}, \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})$. We know that

$$\mathbb{E}[e^{\mathbf{t}^\top \mathbf{X}} | \mathbf{t}^\top \mathbf{X} < K_n] = e^{\mathbf{t}^\top \boldsymbol{\mu} + \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t} / 2} \frac{\Phi\left(\frac{K_n - \mathbf{t}^\top \boldsymbol{\mu}}{\sqrt{\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}}} - \sqrt{\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}}\right)}{\Phi\left(\frac{K_n - \mathbf{t}^\top \boldsymbol{\mu}}{\sqrt{\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}}}\right)}.$$

Taking the derivative with respect to t_i gives

$$\begin{aligned}
\mathbb{E}[X_i e^{\mathbf{t}^\top \mathbf{X}} | \mathbf{t}^\top \mathbf{X} < K_n] &= \left(\mu_i + \sum_{j=1}^n t_j \text{Cov}(X_i, X_j) \right) e^{\mathbf{t}^\top \boldsymbol{\mu} + \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t} / 2} \frac{\Phi\left(\frac{K_n - \mathbf{t}^\top \boldsymbol{\mu}}{\sqrt{\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}}} - \sqrt{\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}}\right)}{\Phi\left(\frac{K_n - \mathbf{t}^\top \boldsymbol{\mu}}{\sqrt{\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}}}\right)} \\
&+ e^{\mathbf{t}^\top \boldsymbol{\mu} + \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t} / 2} \left(\frac{-\mu_i(\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}) - (K_n - \mathbf{t}^\top \boldsymbol{\mu}) \sum_{j=1}^n t_j \text{Cov}(X_i, X_j)}{(\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})^{3/2}} - \frac{\sum_{j=1}^n t_j \text{Cov}(X_i, X_j)}{2\sqrt{\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}}} \right) \frac{\phi\left(\frac{K_n - \mathbf{t}^\top \boldsymbol{\mu}}{\sqrt{\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}}} - \sqrt{\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}}\right)}{\Phi\left(\frac{K_n - \mathbf{t}^\top \boldsymbol{\mu}}{\sqrt{\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}}}\right)} \\
&+ e^{\mathbf{t}^\top \boldsymbol{\mu} + \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t} / 2} \Phi\left(\frac{K_n - \mathbf{t}^\top \boldsymbol{\mu}}{\sqrt{\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}}} - \sqrt{\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}}\right) \left(\frac{-\mu_i(\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}) - (K_n - \mathbf{t}^\top \boldsymbol{\mu}) \sum_{j=1}^n t_j \text{Cov}(X_i, X_j)}{(\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})^{3/2}} \right) \frac{\phi\left(\frac{K_n - \mathbf{t}^\top \boldsymbol{\mu}}{\sqrt{\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}}}\right)}{\left(\Phi\left(\frac{K_n - \mathbf{t}^\top \boldsymbol{\mu}}{\sqrt{\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}}}\right)\right)^2}
\end{aligned}$$

Taking $\mathbf{t}^*$ be the vector of ones gives $(\mathbf{t}^*)^\top \mathbf{X} = S_n \sim \text{Normal}(m_n, s_n^2)$. Since $K_n = m_n + \beta s_n$,

$$\begin{aligned}
\mathbb{E}[X_i e^{S_n} | S_n < K_n] &= \left(\mu_i + \sum_{j=1}^n \text{Cov}(X_i, X_j) \right) e^{m_n + s_n^2 / 2} \frac{\Phi(\beta - s_n)}{\Phi(\beta)} \\
&+ e^{m_n + s_n^2 / 2} \left(\frac{-\mu_i s_n - s_n \beta \sum_{j=1}^n \text{Cov}(X_i, X_j)}{s_n^{1/2}} - \frac{\sum_{j=1}^n \text{Cov}(X_i, X_j)}{2s_n} \right) \frac{\phi(\beta - s_n)}{\Phi(\beta)} \\
&+ e^{m_n + s_n^2 / 2} \Phi(\beta - s_n) \left(\frac{-\mu_i - \beta \sum_{j=1}^n \text{Cov}(X_i, X_j)}{(s_n)^{1/2}} \right) \frac{\phi(\beta)}{(\Phi(\beta))^2}
\end{aligned}$$

Similarly, taking $\mathbf{t}^*$ be the vector of ones divided by Δ_n gives $(\mathbf{t}^*)^\top \mathbf{X} = S_n/\Delta_n$. This gives the desired expression.

3. We have

$$\bar{\mathbb{E}}_n[S_n] = \frac{\mathbb{E}[S_n e^{\bar{S}_n/\Delta_n}]}{\mathbb{E}[e^{\bar{S}_n/\Delta_n}]},$$

where

$$\mathbb{E}[S_n e^{\bar{S}_n/\Delta_n}] = \mathbb{E}[S_n e^{S_n/\Delta_n} | S_n < K_n] \mathbb{P}[S_n < K_n] + \mathbb{E}[S_n e^{K_n/\Delta_n} | S_n \geq K_n] \mathbb{P}[S_n \geq K_n].$$

All other things are known except $\mathbb{E}[S_n e^{S_n/\Delta_n} | S_n < K_n]$. Note that for the normal random variable $X \sim \text{Normal}(\mu, \sigma^2)$, the moment generating function of its truncation $X < b$ is given by

$$\mathbb{E}[e^{tX} | X < b] = e^{\mu t + \sigma^2 t^2/2} \frac{\Phi\left(\frac{b-\mu}{\sigma} - \sigma t\right)}{\Phi\left(\frac{b-\mu}{\sigma}\right)}.$$

Taking a derivative on both sides with respect to t gives

$$\mathbb{E}[X e^{tX} | X < b] = \frac{e^{\mu t + \sigma^2 t^2/2}}{\Phi\left(\frac{b-\mu}{\sigma}\right)} \left((\mu + \sigma^2 t) \Phi\left(\frac{b-\mu}{\sigma} - \sigma t\right) - \sigma \phi\left(\frac{b-\mu}{\sigma} - \sigma t\right) \right).$$

This implies that

$$\mathbb{E}[S_n e^{S_n/\Delta_n} | S_n < K_n] = \frac{e^{\frac{m_n}{\Delta_n} + \frac{s_n^2}{2\Delta_n^2}}}{\Phi(\beta)} \left(\left(m_n + \frac{s_n^2}{\Delta_n} \right) \Phi\left(\beta - \frac{s_n}{\Delta_n}\right) - s_n \phi\left(\beta - \frac{s_n}{\Delta_n}\right) \right).$$

Therefore,

$$\bar{\mathbb{E}}_n[S_n] = \frac{e^{\frac{m_n}{\Delta_n} + \frac{s_n^2}{2\Delta_n^2}} \left(\left(m_n + \frac{s_n^2}{\Delta_n} \right) \Phi\left(\beta - \frac{s_n}{\Delta_n}\right) - s_n \phi\left(\beta - \frac{s_n}{\Delta_n}\right) \right) + e^{\frac{(m_n + s_n \beta)}{\Delta_n}} \left(m_n + \frac{s_n \phi(\beta)}{1 - \Phi(\beta)} \right) (1 - \Phi(\beta))}{\left(e^{\frac{1}{\Delta_n} m_n + \frac{1}{2\Delta_n^2} s_n^2} \right) \Phi\left(\beta - \frac{1}{\Delta_n} s_n\right) + e^{(m_n + s_n \beta)/\Delta_n} (1 - \Phi(\beta))}$$

4. Stop loss transformation has known formula.

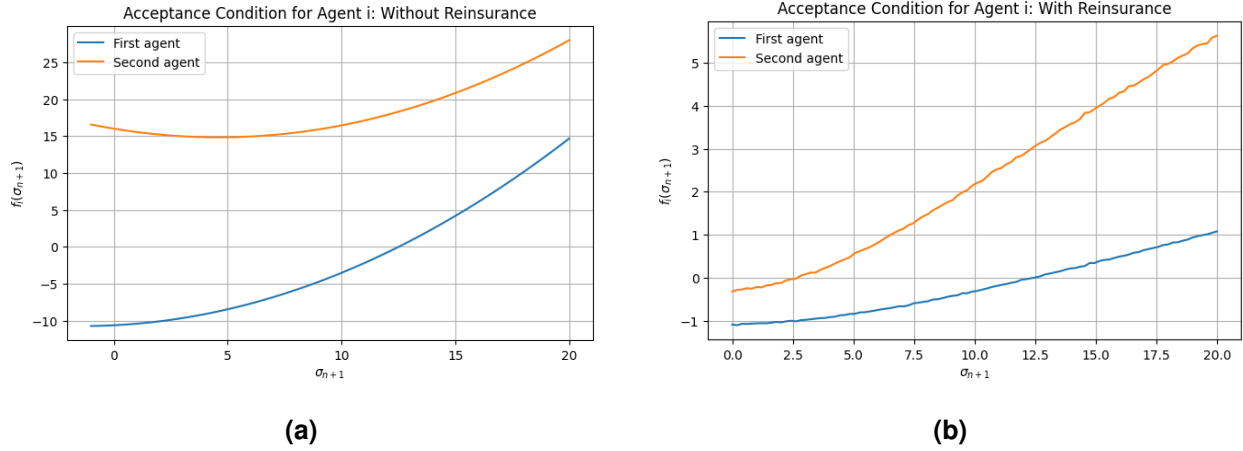
$$\begin{aligned} \mathbb{E}[(S_n - K_n)_+] &= \mathbb{E}[S_n - K_n | S_n \geq K_n] \mathbb{P}[S_n \geq K_n] \\ &= s_n (\phi(\beta) - \beta(1 - \Phi(\beta))). \end{aligned}$$

Thus, we have the closed form formula for $\rho_i(Y_i^{(n)}) \geq \rho_i(Y_i^{(n+1)})$.

G.2. Example: the effect of reinsurance on the expansion under EP

We present a simplified example that highlights the effect of reinsurance. Consider a risk pool with only two members, i.e. $n = 2$ where $X_1 \sim \text{Normal}(5, 9)$, $X_2 \sim \text{Normal}(9, 169)$, $\rho_{X_1, X_2} = 0.2$, $\delta_1 = \delta_2 = 1$, and $K = 25.4$ ($\beta = 0.7125$). Observe that the second existing member has a much higher risk. We first consider the effect of reinsurance on existing members. We readily get that

$$\rho_1(\bar{Y}_2) \approx 2.92 > -2.87 \approx \rho_1(Y_2),$$

Figure 11 Example G.2.

which means that the first member's position worsens due to the reinsurance. We have seen in Remark 6 that the aggregate gain from the reinsurance is positive *i.e.* $\rho_1(Y_2) + \rho_2(Y_2) > \rho_1(\bar{Y}_2) + \rho_2(\bar{Y}_2)$, which implies that the second member benefits from the reinsurance (as expected). How is the situation affected by a new entrance? Without reinsurance, the gain of the second agent from the pool's expansion is much higher than that of a first agent when σ_3 is large (see Figure 11a). With reinsurance, benefits from the pool's expansion are lower for both members, and acceptance regions (in terms of σ_3) become narrower for both members (see Figure 11b). Comparing Figures 11a and 11b, we note that the second agent no longer always accepts the candidate in the presence of reinsurance and that the first agent has a slightly higher requirement for the variance of the candidate. This example is consistent with the fact that reinsurance may reduce gains from a new entrance and hence existing members impose stricter requirements for accepting a candidate.

G.3. Proofs of Section 6

Proof of Theorem 14

The following inequalities are equivalent:

$$\begin{aligned}
 \bar{\mathbb{E}}[(S_n - K)_+] &\geq \mathbb{E}[(S_n - K)_+] \\
 \frac{\mathbb{E}[(S_n - K)_+ e^{\bar{S}_n/\Delta_n}]}{\mathbb{E}[e^{\bar{S}_n/\Delta_n}]} &\geq \mathbb{E}[(S_n - K)_+] \\
 \mathbb{E}[(S_n - K)_+ e^{\bar{S}_n/\Delta_n}] &\geq \mathbb{E}[(S_n - K)_+] \mathbb{E}[e^{\bar{S}_n/\Delta_n}] \\
 \mathbb{E}[(S_n - K)_+ e^{\bar{S}_n/\Delta_n}] &\geq \mathbb{E}[(S_n - K)_+] \mathbb{E}[e^{\bar{S}_n/\Delta_n}] \\
 \text{Cov}((S_n - K)_+, e^{\bar{S}_n/\Delta_n}) &\geq 0.
 \end{aligned}$$

The latter always holds because $(S_n - K)_+$ and $e^{\bar{S}_n/\Delta_n}$ are non-decreasing functions of S_n .

Proof of Theorem 15

For the stability we need to verify that the risk measure of each individual agent under risk-sharing with reinsurance is lower than the corresponding when he is outside the pool. This however comes from the definition of the demand function $\bar{z}_i^*$. Indeed, at equilibrium prices $(\mathbf{p}^*, \bar{p}^*)$, the optimal demand is different than zero and hence the participation in the equilibrium risk sharing is better than staying outside (which is equivalent to zero demand). Taking into account the fact that under the suggested cash transfer (21) the payment is lower than that at the equilibrium we get the verification of the stability for each agent.

The argument is similar for the weak consensus, since therein we compare the risk reduction inside a pool of $n + 1$ agents and the staying outside of risk-sharing.

Proof of Remark 6

We need to show the following inequality:

$$\sum_{i=1}^n \rho_i(X_i + a^* \cdot X + a_i^* \cdot p^*) \geq \sum_{i=1}^n \rho_i(X_i + \bar{a}_i \cdot X + \bar{a}_i \bar{S}_n - \bar{a}_i \cdot \bar{p}^* - \bar{a}_i \bar{p}) \quad (12)$$

where the (a^*, p^*) are the equilibrium allocation and prices without reinsurance and $(\bar{a}, \bar{p})$ the corresponding quantities at the equilibrium with reinsurance at level K . From the calculation of both equilibria, we have that inequality is equivalent to

$$\Delta \log \left(\mathbb{E}[e^{\bar{S}_n/\Delta}] \right) + \mathbb{E}[(S_n - K)_+] \leq \Delta \log \left(\mathbb{E}[e^{S_n/\Delta}] \right). \quad (13)$$

This is equivalent to

$$\begin{aligned} 0 &\leq \log \left(\frac{\mathbb{E}[e^{S_n/\Delta}]}{\mathbb{E}[e^{\bar{S}_n/\Delta}]} \right) - \frac{1}{\Delta} \mathbb{E}[(S_n - K)_+] \\ 1 &\leq \frac{\mathbb{E}[e^{S_n/\Delta}]}{\mathbb{E}[e^{\bar{S}_n/\Delta}] e^{\mathbb{E}[(S_n - K)_+]/\Delta}} \\ \mathbb{E}[e^{\bar{S}_n/\Delta}] e^{\mathbb{E}[(S_n - K)_+]/\Delta} &\leq \mathbb{E}[e^{S_n/\Delta}]. \end{aligned}$$

By the Jensen inequality we have that $e^{\mathbb{E}[(S_n - K)_+]/\Delta} \leq \mathbb{E}[e^{(S_n - K)_+/\Delta}]$ and since $e^{(S_n - K)_+}$ and $e^{\bar{S}_n/\Delta}$ have positive Covariance (they are both monotonic to S_n), we have that

$$\mathbb{E}[e^{\bar{S}_n/\Delta}] e^{\mathbb{E}[(S_n - K)_+]/\Delta} \leq \mathbb{E}[e^{\bar{S}_n/\Delta}] \mathbb{E}[e^{(S_n - K)_+/\Delta}] \leq \mathbb{E}[e^{\bar{S}_n/\Delta} e^{(S_n - K)_+/\Delta}] = \mathbb{E}[e^{S_n/\Delta}],$$

which completes the proof.

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